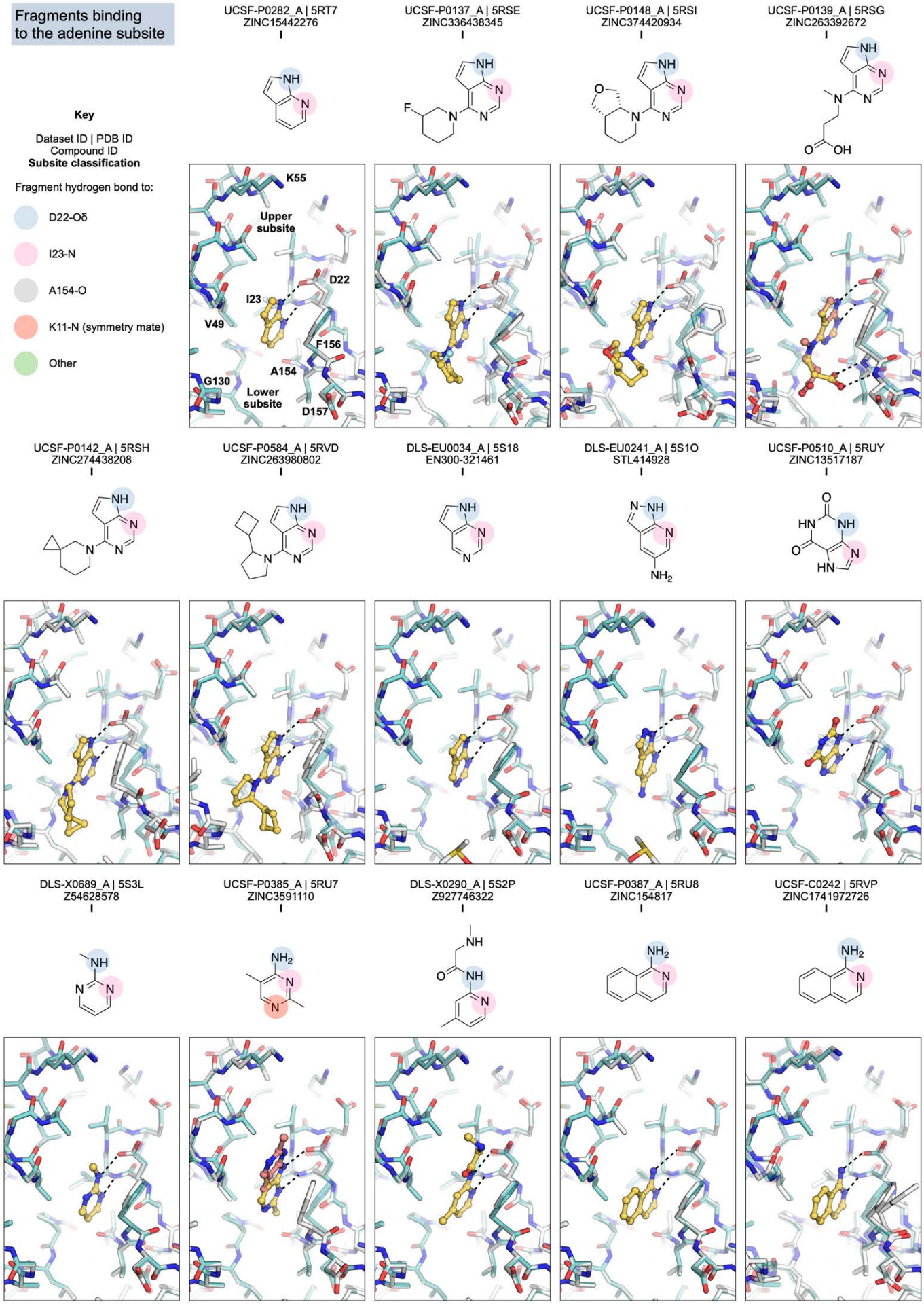
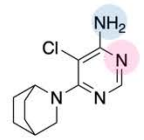
Data S2

Structures of fragments bound to Mac1

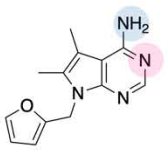
Only fragments binding in the adenosine site, catalytic site or K90 site are shown. Fragments are shown as yellow sticks. Mac1 is shown in the fragment-bound state (white sticks) and the apo state (transparent teal sticks). Hydrogen bonds are shown as dashed black lines. The Mac1 subsites are defined in Figure 2 of the main text. The dataset code, PDB code and compound ID are given for each fragment, along with motif classification for fragments binding to the adenine and oxyanion subsites.


Fragments binding to the adenine subsite
Key
Dataset ID | PDB ID
Compound ID
Subsite classification
Fragment hydrogen bond to:
D22-Oδ
I23-N
A154-O
K11-N (symmetry mate)
Other
UCSF-P0282_A | 5RT7
ZINC15442276
UCSF-P0137_A | 5RSE
ZINC336438345
UCSF-P0148_A | 5RSI
ZINC374420934
UCSF-P0139_A | 5RSG
ZINC263392672
K55
Upper subsite
D22
I23
V49
F156
A154
G130
Lower subsite
D157
UCSF-P0142_A | 5RSH
ZINC274438208
UCSF-P0584_A | 5RVD
ZINC263980802
DLS-EU0034_A | 5S18
EN300-321461
DLS-EU0241_A | 5S1O
STL414928
UCSF-P0510_A | 5RUY
ZINC13517187
DLS-X0689_A | 5S3L
Z54628578
UCSF-P0385_A | 5RU7
ZINC3591110
DLS-X0290_A | 5S2P
Z927746322
UCSF-P0387_A | 5RU8
ZINC154817
UCSF-C0242 | 5RVP
ZINC1741972726

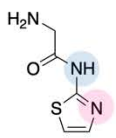
UCSF-P0163_A | 5RSL
ZINC365052868
I



UCSF-P0131_A | 5RSC
ZINC3888754
I



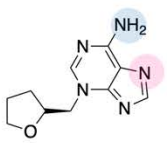
DLS-EU0238_A | 5S1M
STK497968
I

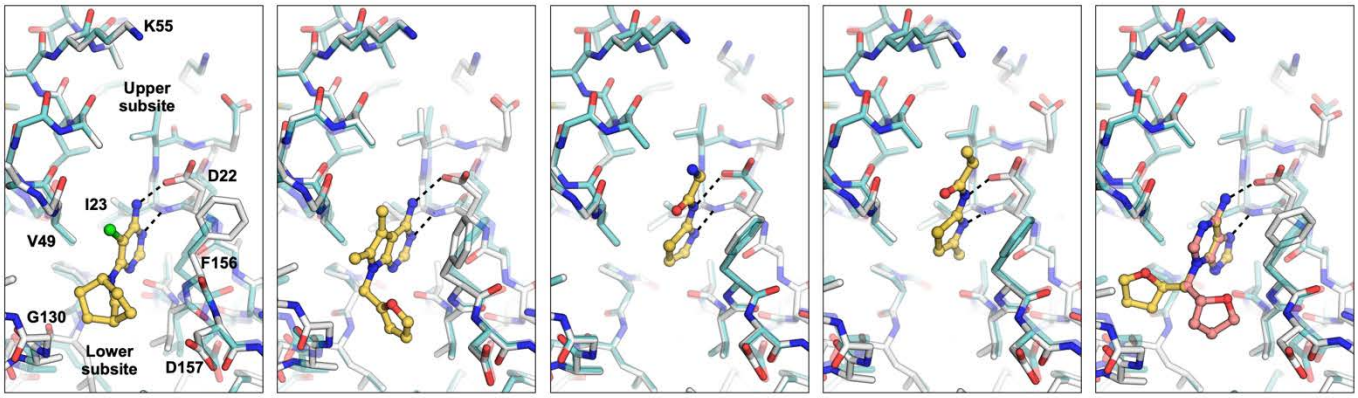


DLS-X0626_A | 5S3D
Z30820160
I

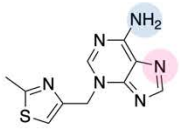


UCSF-P0588_A | 5RVF
ZINC82473428_N3
II

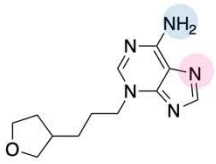




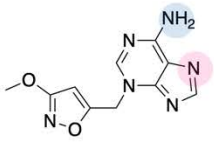
UCSF-P0159_A | 5RSJ
ZINC89254160_N3
II



UCSF-P0590_A | 5RVG
ZINC400552187_N3
II



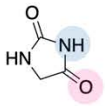
UCSF-P0161_A | 5RSK
ZINC901381520_N3
II

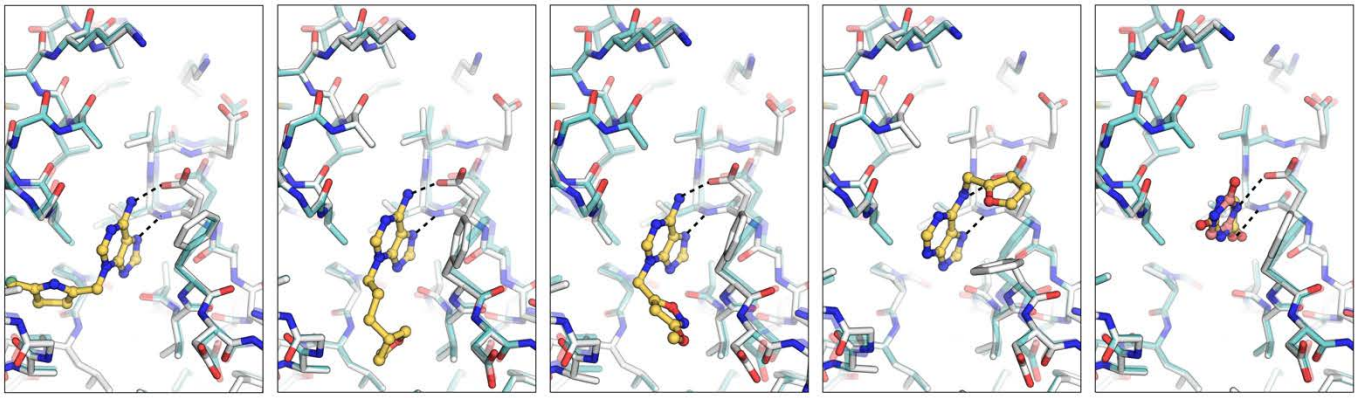


UCSF-P0116_A | 5RS8
ZINC1601
II

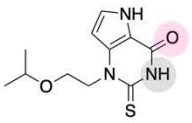


DLS-X0895_A | 5S46
Z57131035
III





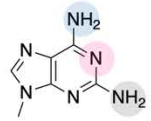
UCSF-P0115_A | 5RS7
ZINC34618676
IV



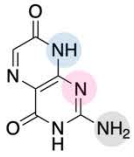
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V



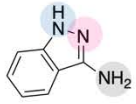
UCSF-P0138_A | 5RSF
ZINC26180281
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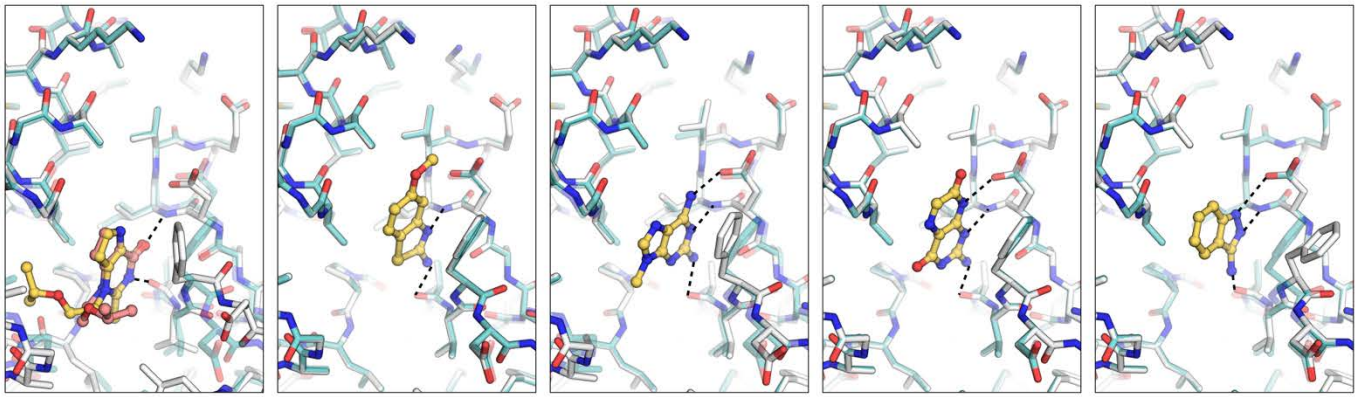


UCSF-P0576_A | 5RVB
ZINC14419577
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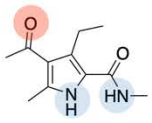


UCSF-P0567_A | 5RV7
ZINC3954002
VII

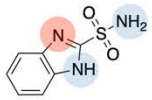




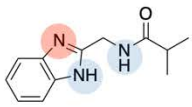
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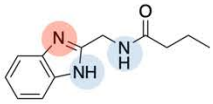
UCSF-P0303_A | 5RTC
ZINC6490906
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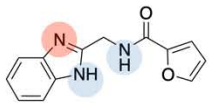
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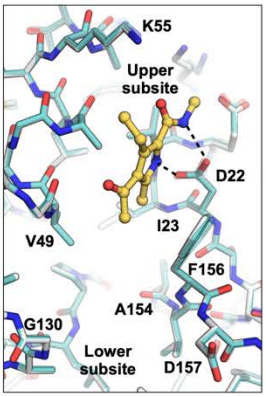


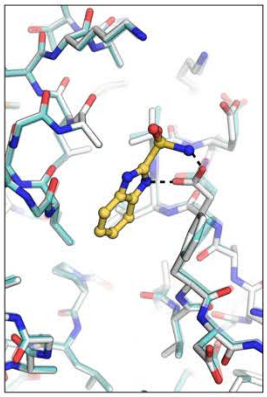
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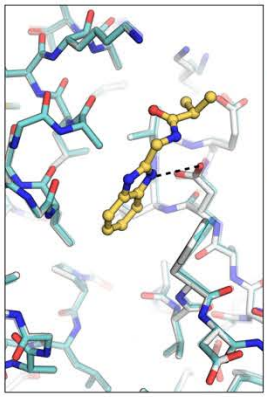


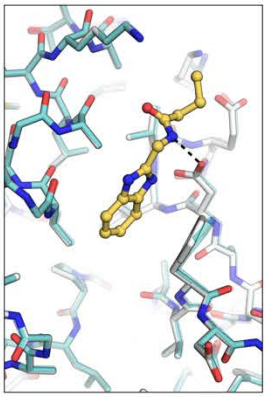
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VIII

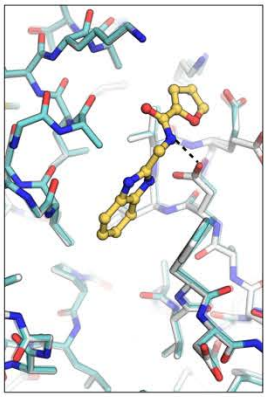




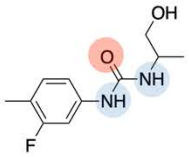




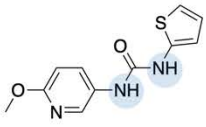




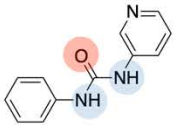
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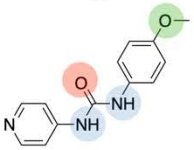
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IX



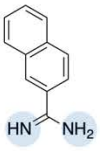
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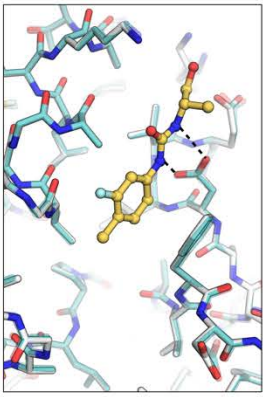


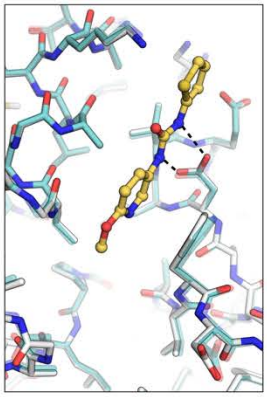
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IX

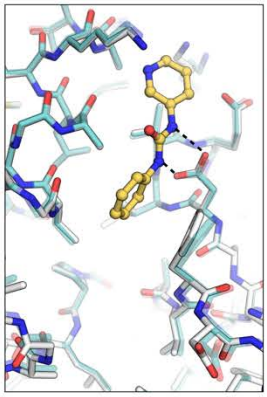


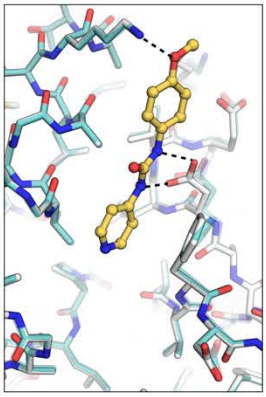
UCSF-P0381_B | 5RU6
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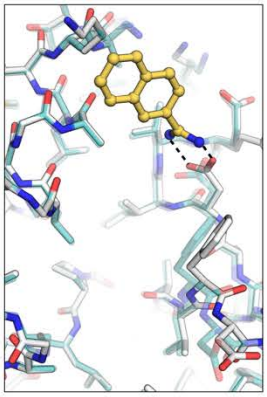




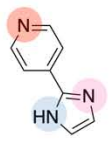








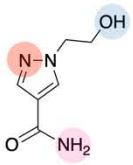
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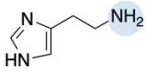
DLS-X0910_A | 5S49
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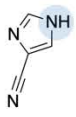
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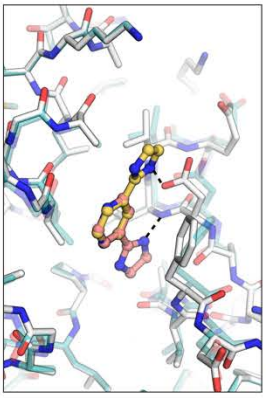


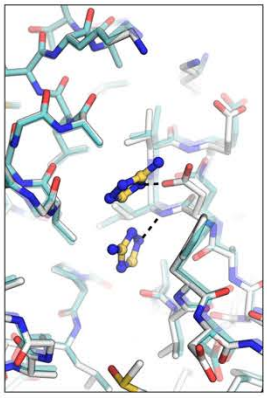
UCSF-P0453_A | 5RUS
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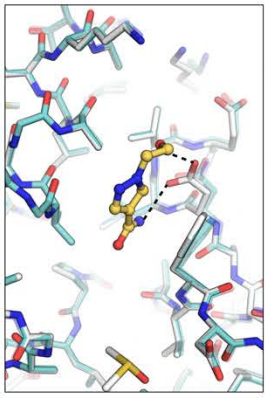


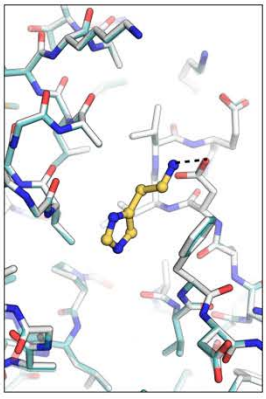
DLS-X0962_A | 5S4E
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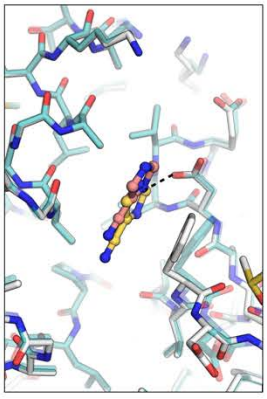




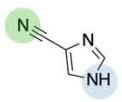




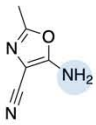




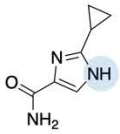
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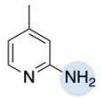
DLS-EU0144_A | 5S1I
EN300-301084



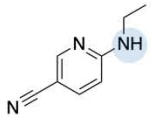
DLS-X0421_A | 5S2W
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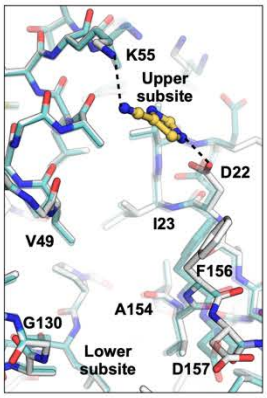


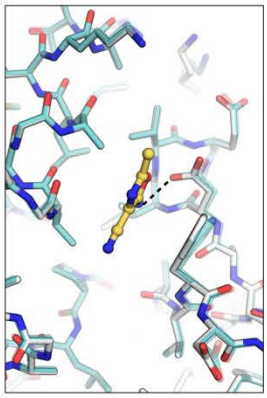
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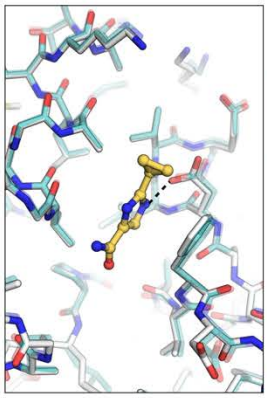


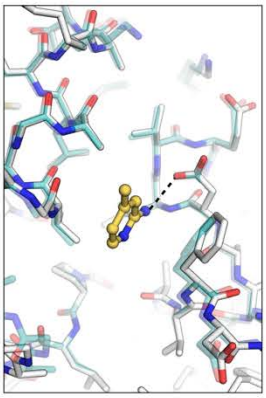
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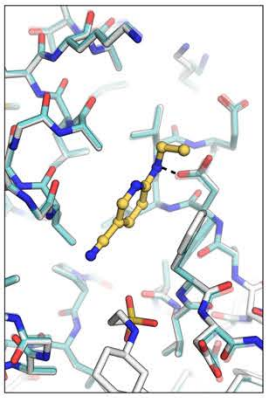








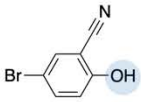




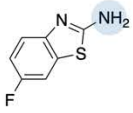
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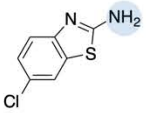
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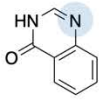
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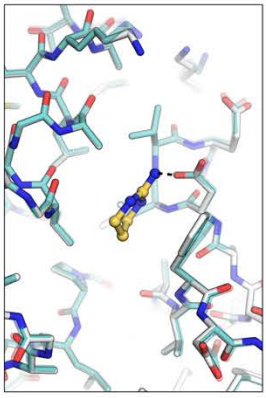


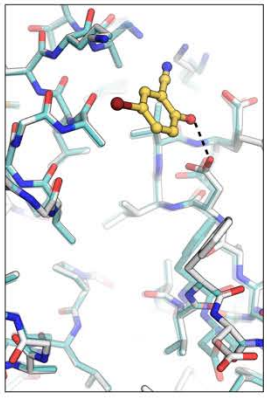
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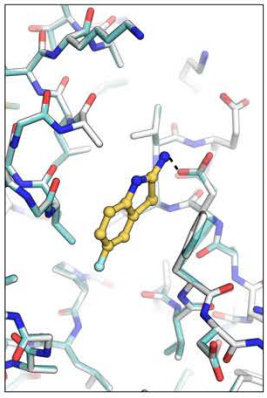


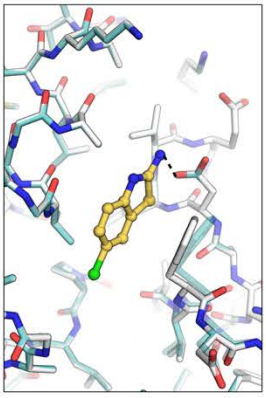
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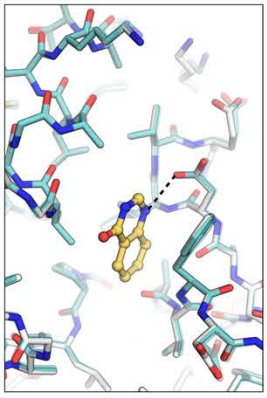




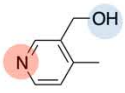




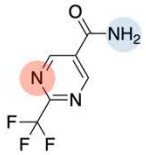




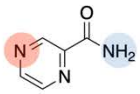
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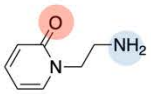
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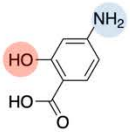
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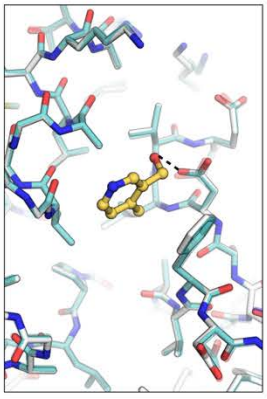


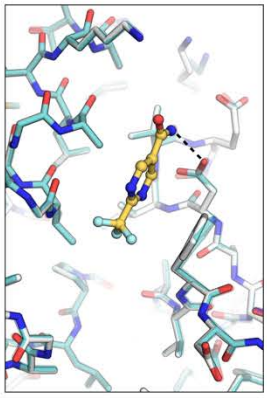
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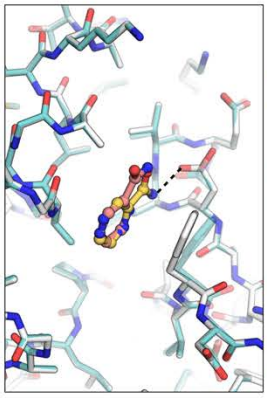


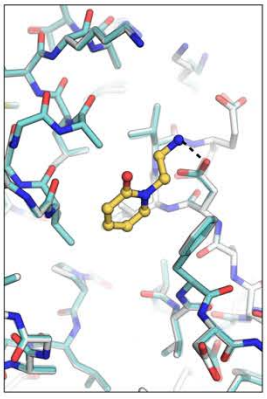
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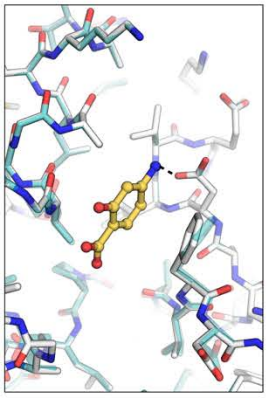




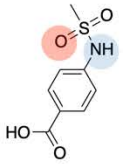




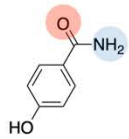




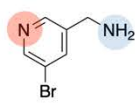
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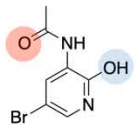
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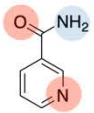
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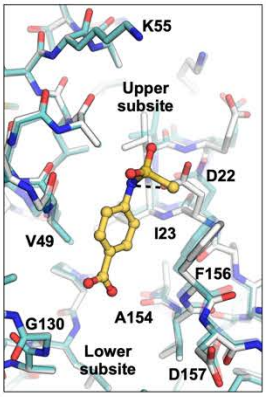


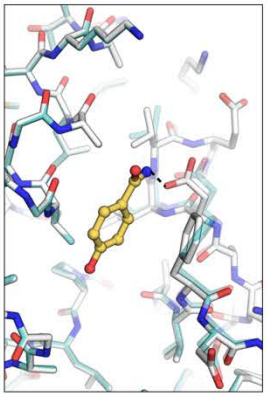
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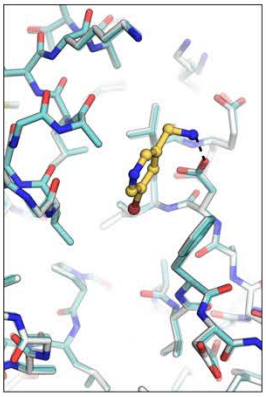


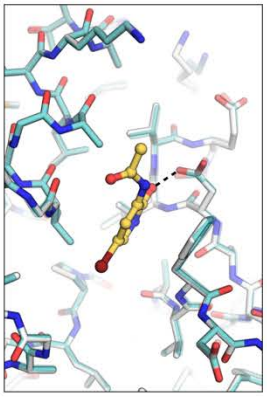
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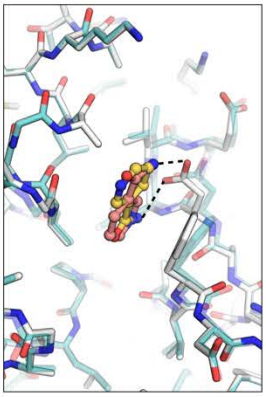




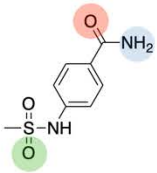




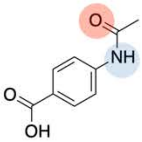




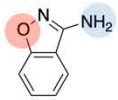
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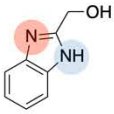
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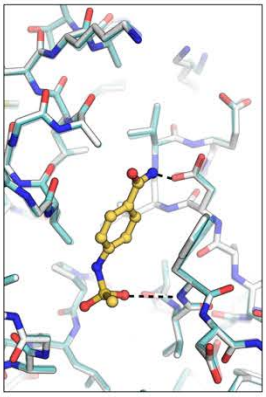


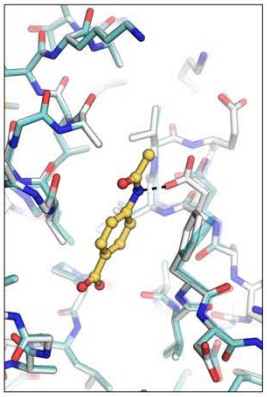
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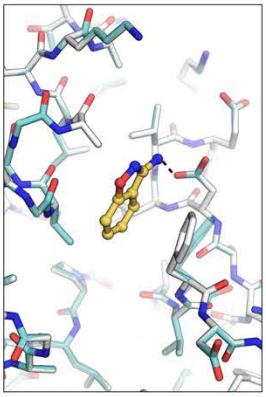


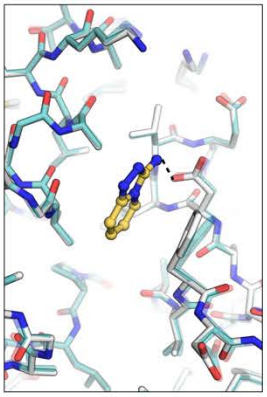
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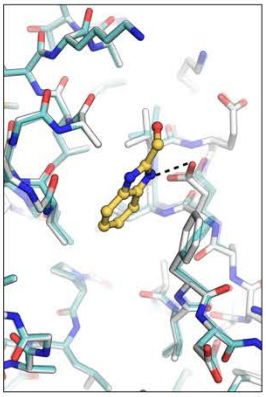




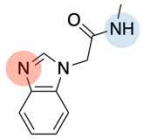




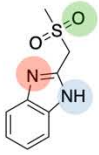




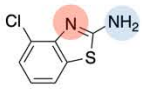
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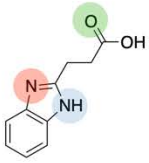
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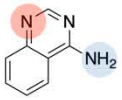
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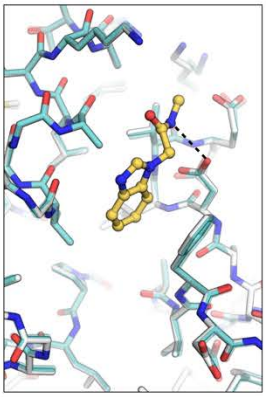


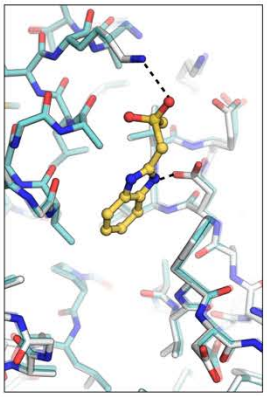
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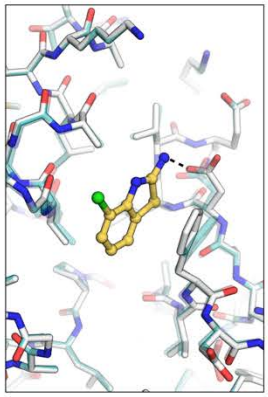


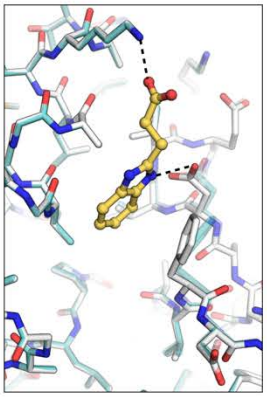
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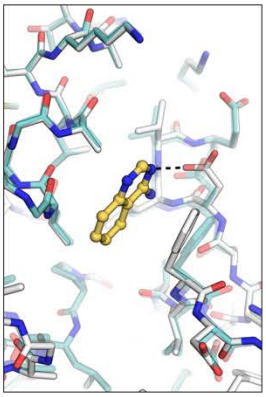




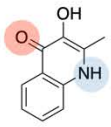








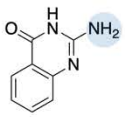
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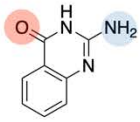
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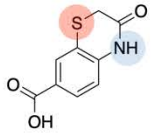
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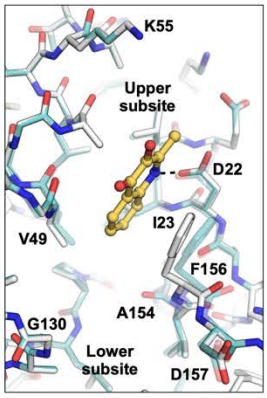


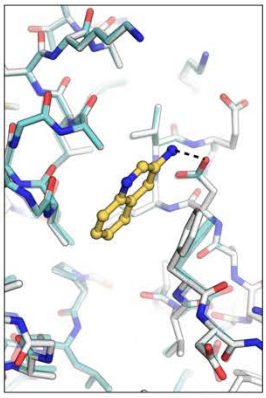
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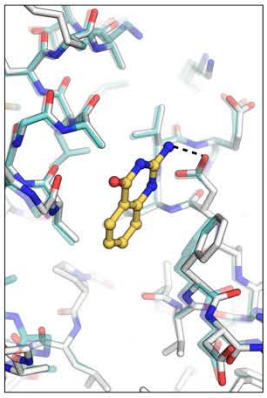


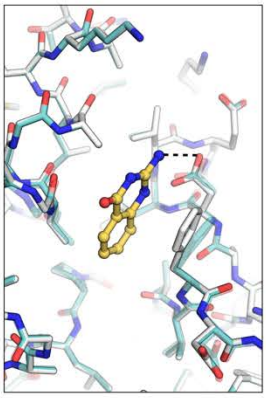
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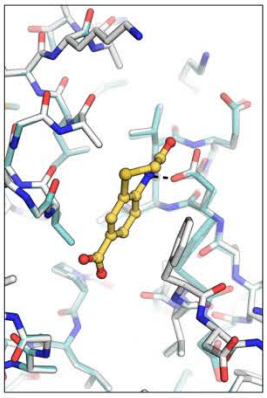








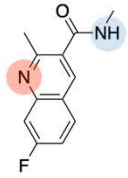




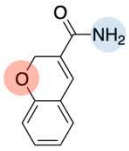
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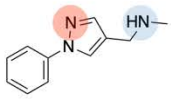
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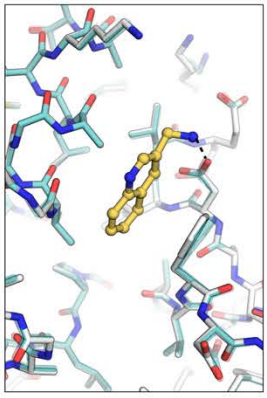


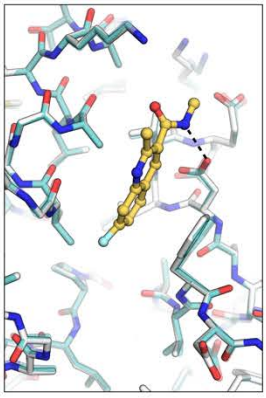
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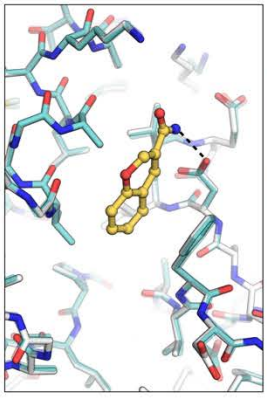


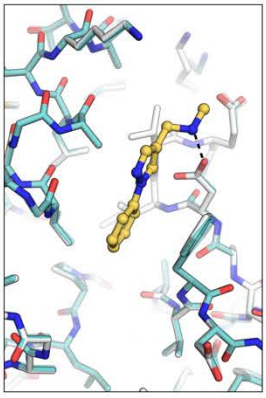
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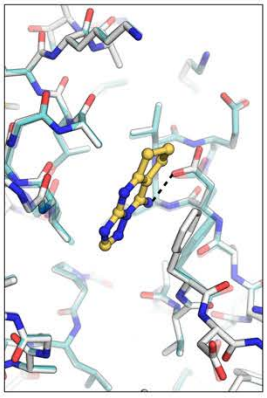




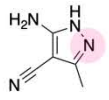




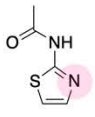




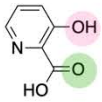
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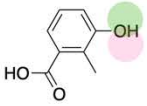
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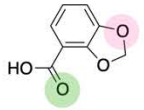
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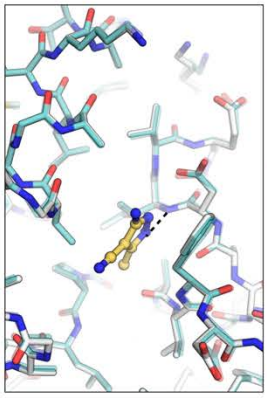


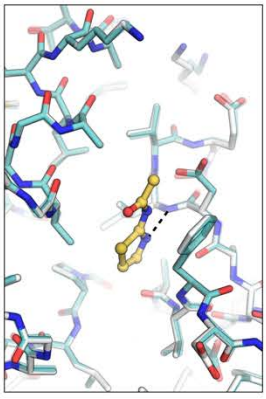
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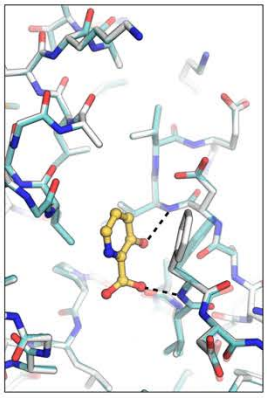


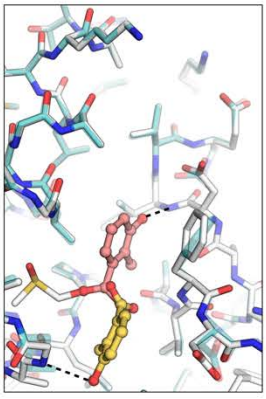
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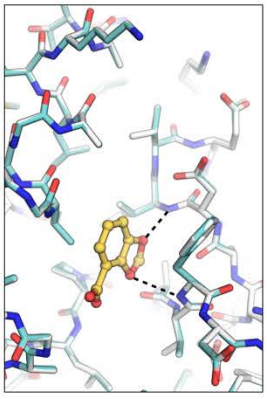




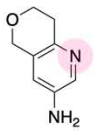








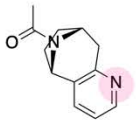
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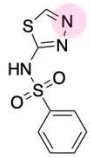
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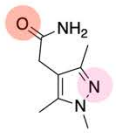
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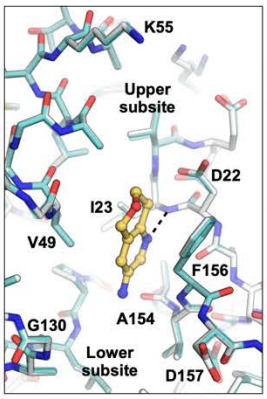


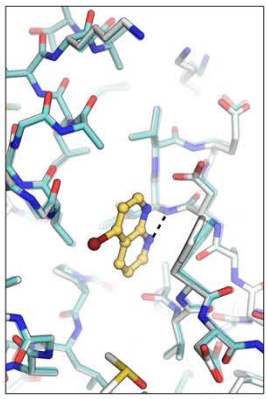
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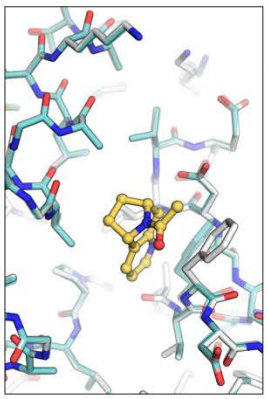


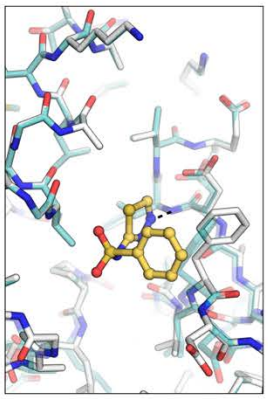
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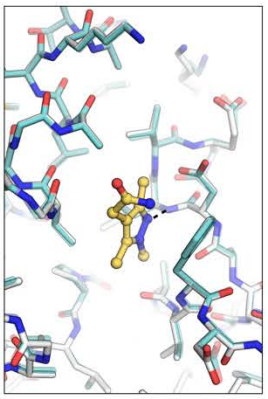




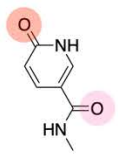








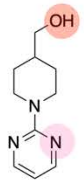
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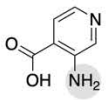
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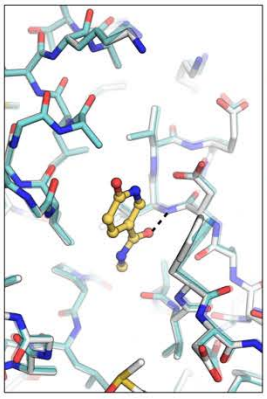


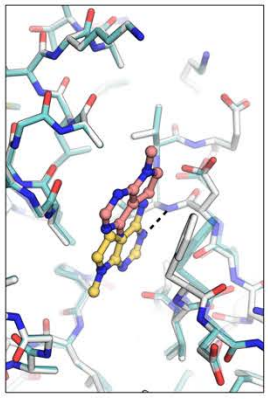
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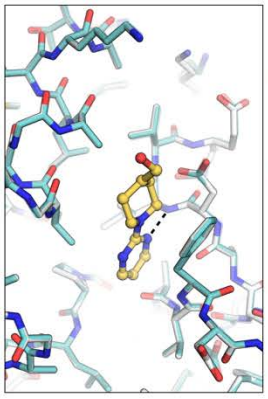


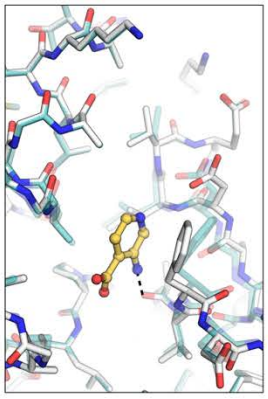
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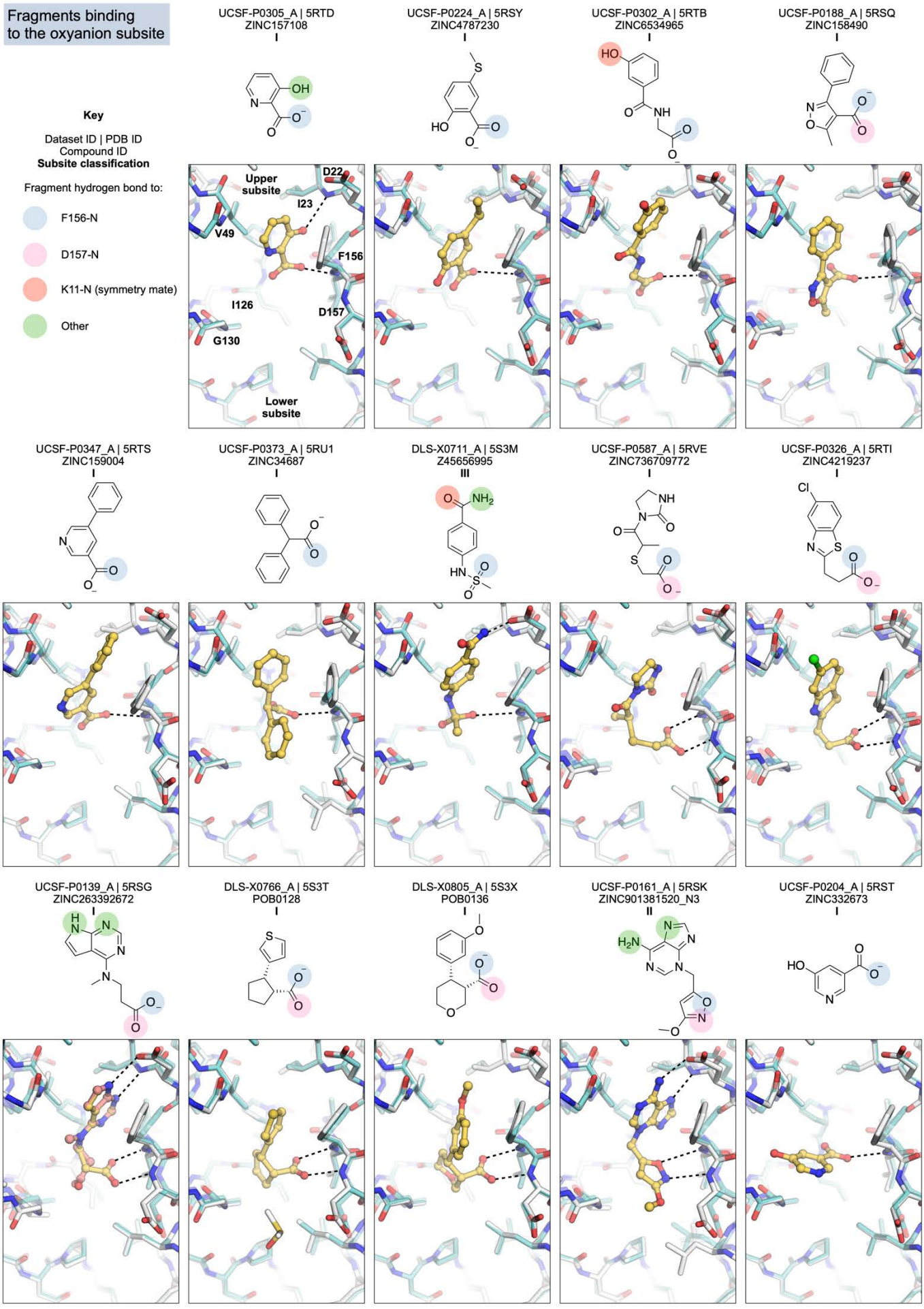




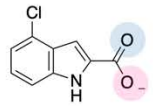







Fragments binding to the oxyanion subsite
Key
Dataset ID | PDB ID
Compound ID
Subsite classification
Fragment hydrogen bond to:
F156-N
D157-N
K11-N (symmetry mate)
Other
UCSF-P0305_A | 5RTD
ZINC157108
UCSF-P0224_A | 5RSY
ZINC4787230
UCSF-P0302_A | 5RTB
ZINC6534965
UCSF-P0188_A | 5RSQ
ZINC158490
Upper subsite
D22
I23
V49
F156
I126
D157
G130
Lower subsite
UCSF-P0347_A | 5RTS
ZINC159004
UCSF-P0373_A | 5RU1
ZINC34687
DLS-X0711_A | 5S3M
Z45656995
III
UCSF-P0587_A | 5RVE
ZINC736709772
UCSF-P0326_A | 5RTI
ZINC4219237
UCSF-P0139_A | 5RSG
ZINC263392672
DLS-X0766_A | 5S3T
POB0128
DLS-X0805_A | 5S3X
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II
UCSF-P0204_A | 5RST
ZINC332673

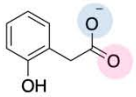
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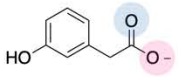
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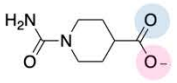
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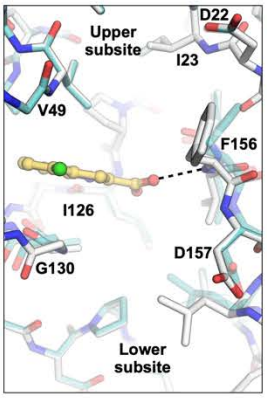


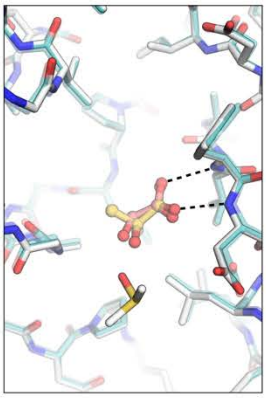
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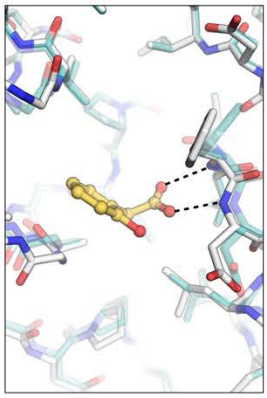


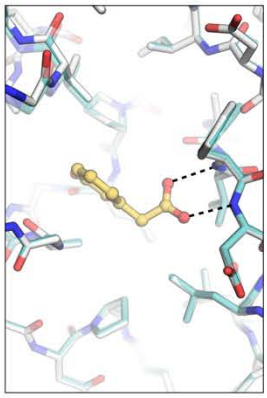
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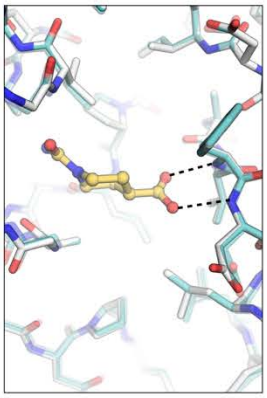




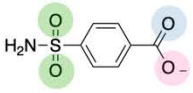




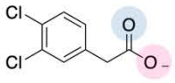




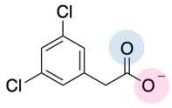
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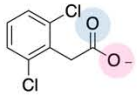
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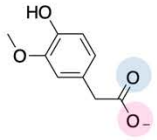
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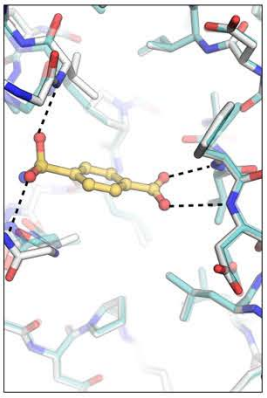


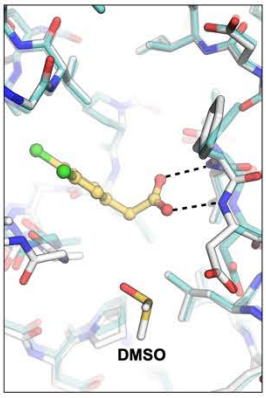
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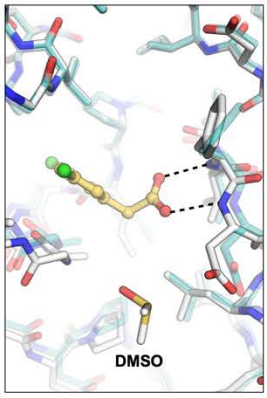


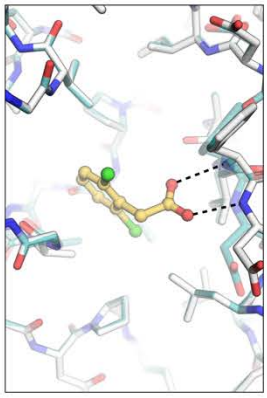
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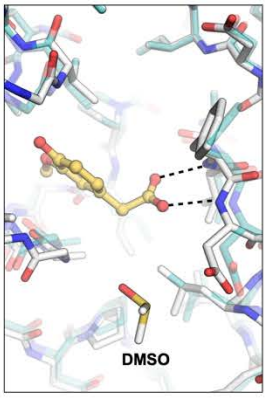




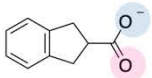




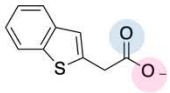




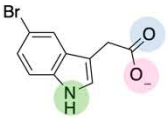
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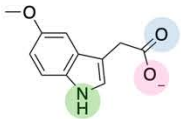
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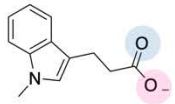
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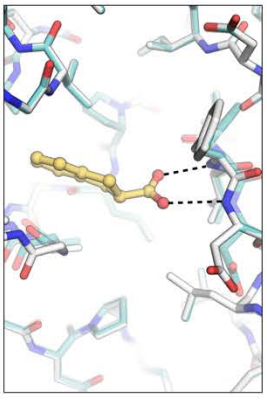


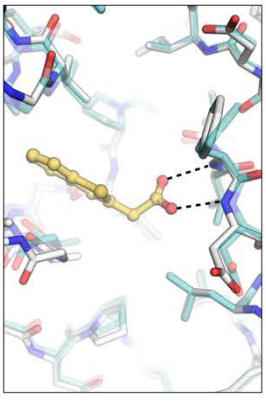
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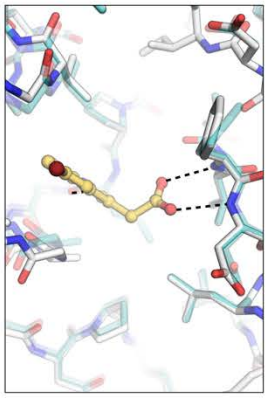


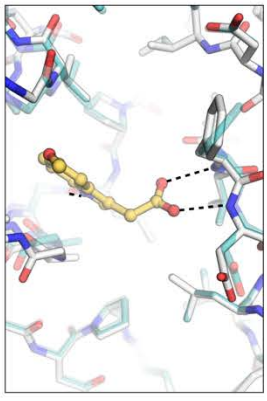
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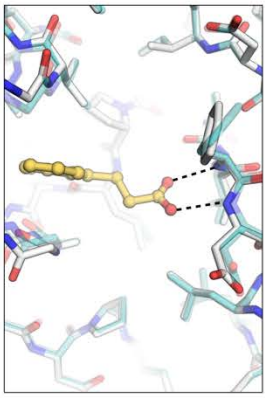


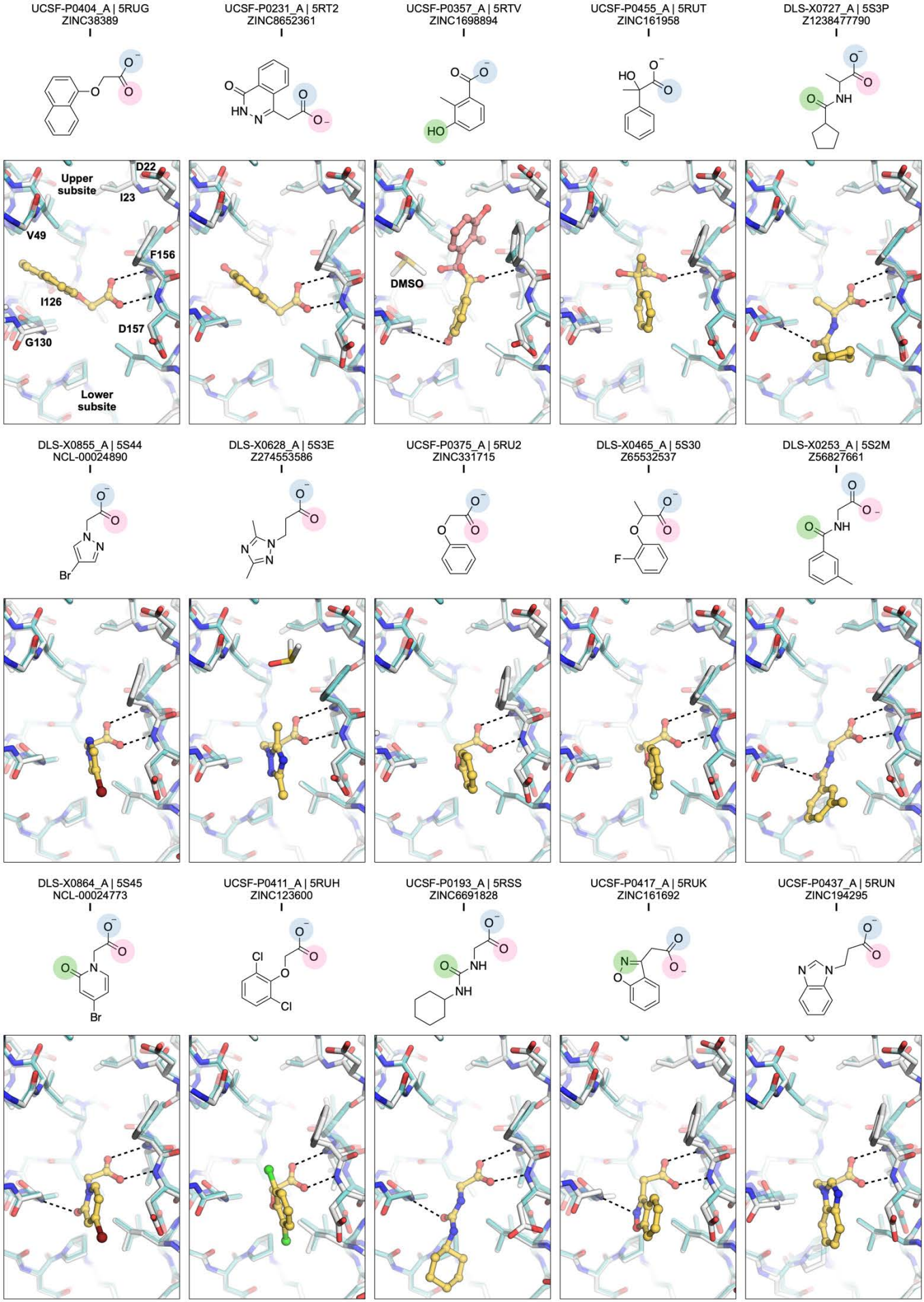


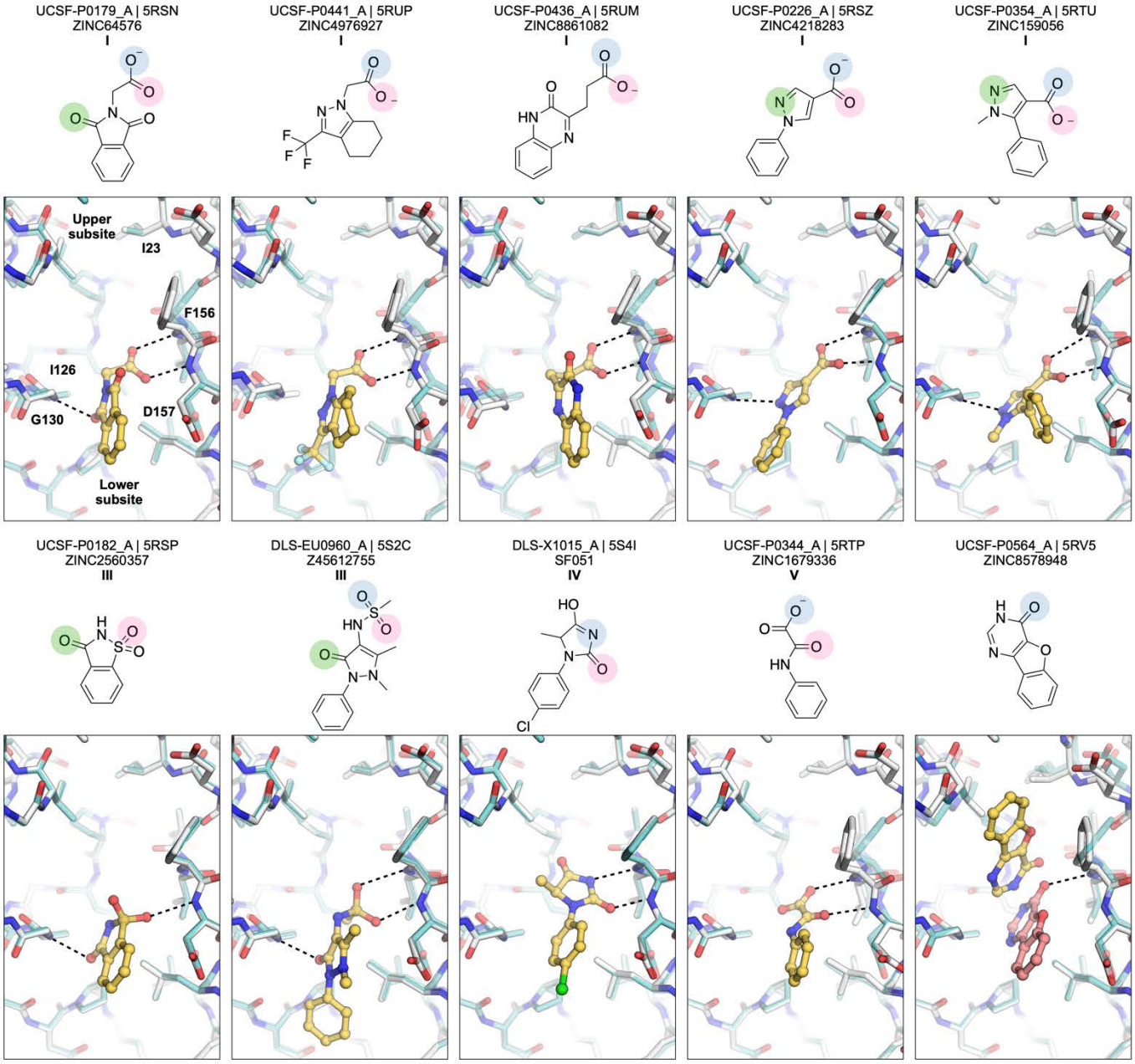









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UCSF-P0231_A | 5RT2
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UCSF-P0357_A | 5RTV
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DLS-X0727_A | 5S3P
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Upper subsite
D22
I23
V49
F156
I126
D157
G130
Lower subsite
DMSO
DLS-X0855_A | 5S44
NCL-00024890
DLS-X0628_A | 5S3E
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UCSF-P0375_A | 5RU2
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DLS-X0253_A | 5S2M
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DLS-X0864_A | 5S45
NCL-00024773
UCSF-P0411_A | 5RUH
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UCSF-P0193_A | 5RSS
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UCSF-P0417_A | 5RUK
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UCSF-P0437_A | 5RUN
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UCSF-P0179_A | 5RSN
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UCSF-P0441_A | 5RUP
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I
UCSF-P0354_A | 5RTU
ZINC159056
I
Upper subsite
I23
F156
I126
G130
D157
Lower subsite
UCSF-P0182_A | 5RSP
ZINC2560357
III
DLS-EU0960_A | 5S2C
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III
DLS-X1015_A | 5S4I
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IV
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Fragments binding to the catalytic site

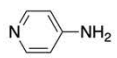
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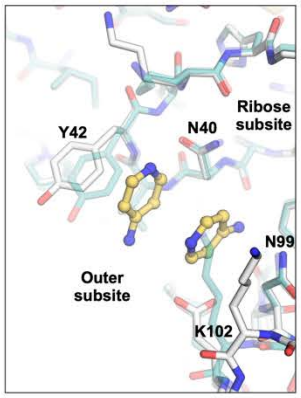
Dataset ID | PDB ID
Compound ID

Fragment-protein interaction:

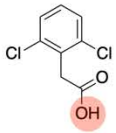
- H-bond
- Salt bridge

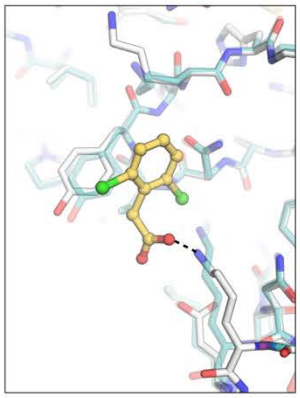
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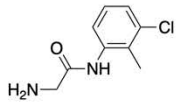


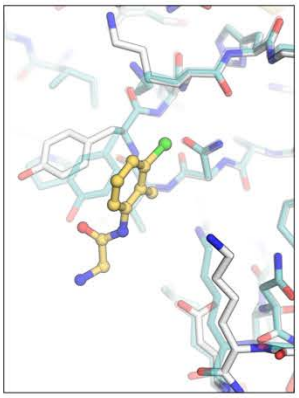
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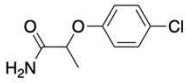


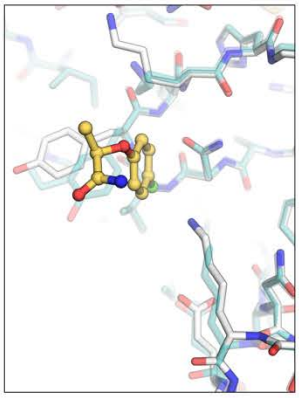
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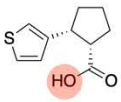


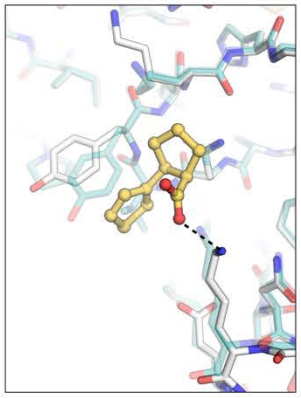
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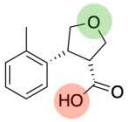


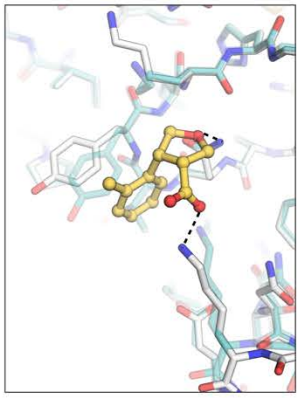
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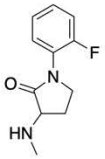


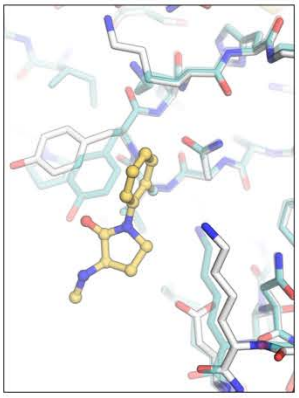
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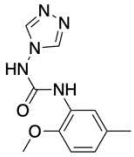


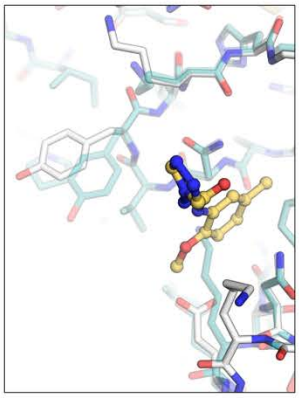
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DLS-X0228_B | 5S2L
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Fragments binding to the K90 site

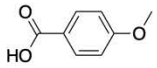
Key

Dataset ID | PDB ID
Compound ID

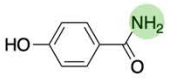
Fragment-protein interaction:

H-bond

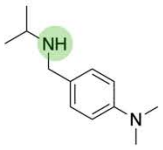
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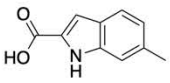
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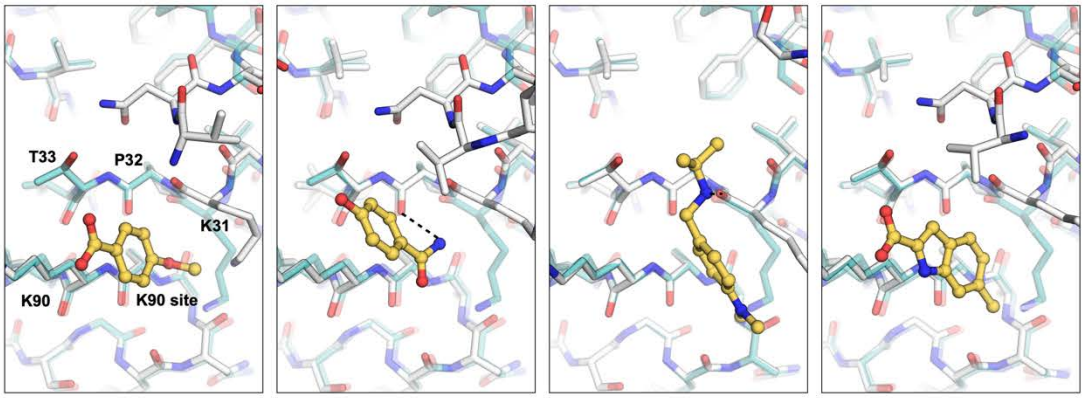


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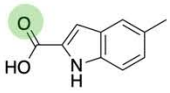


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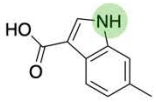




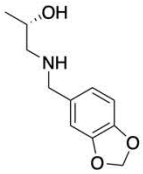
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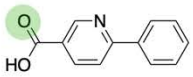
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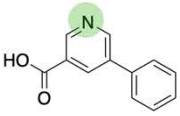
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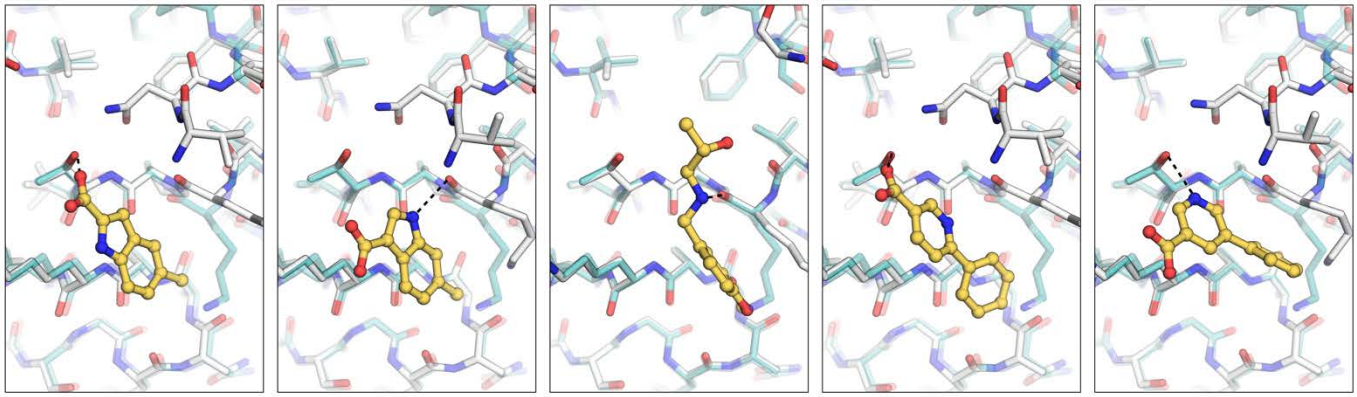


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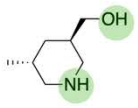


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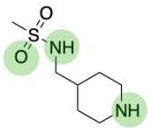




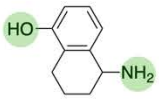
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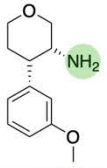
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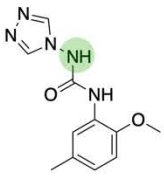
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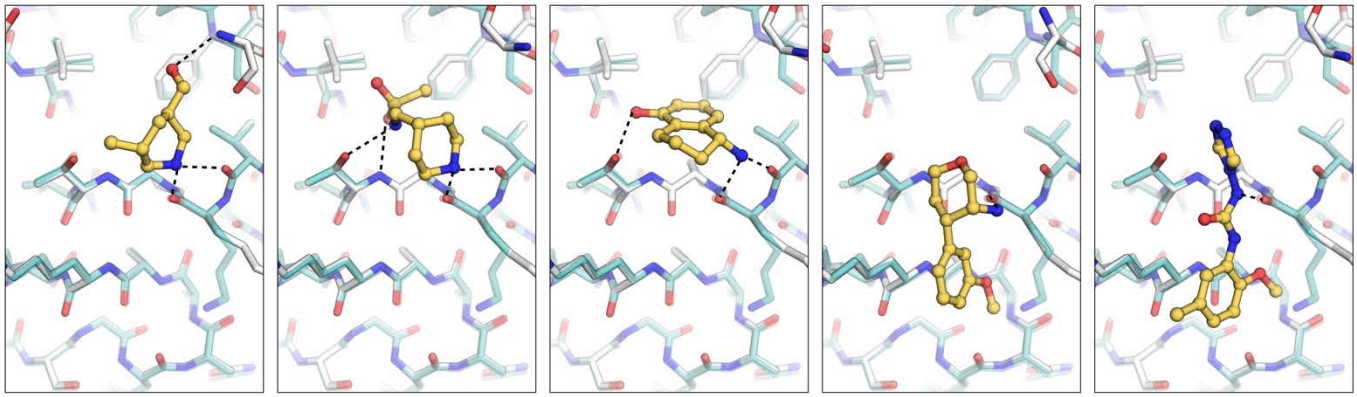


DLS-X0822_A | 5S3Z
POB0140



DLS-X0228_A | 5S2L
Z2234920345





Fragments binding to the adenosine site (not binding to the adenine or oxyanion subsites)

Key

Dataset ID | PDB ID
Compound ID

Fragment hydrogen bond to:

K11-N (symmetry mate)

Other

DLS-X0900_A | 5S47
Z940713508



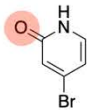
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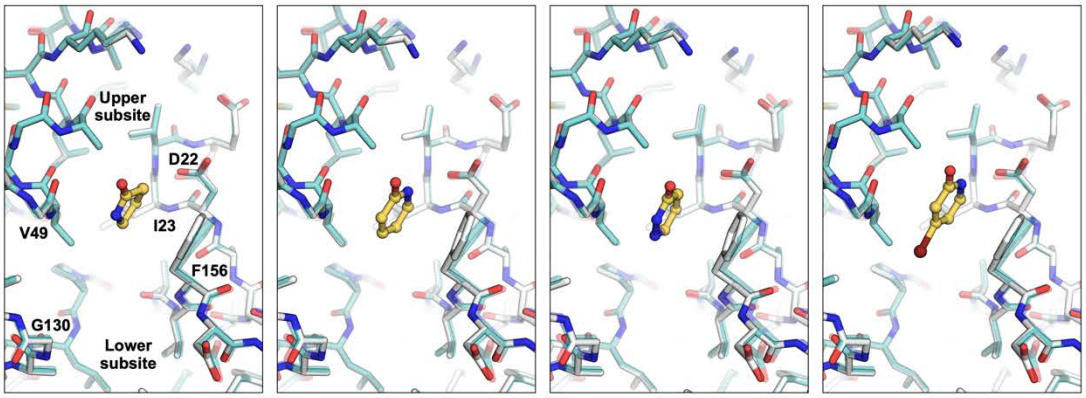


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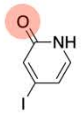


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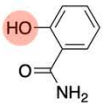




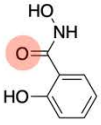
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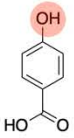
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ZINC2055



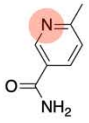
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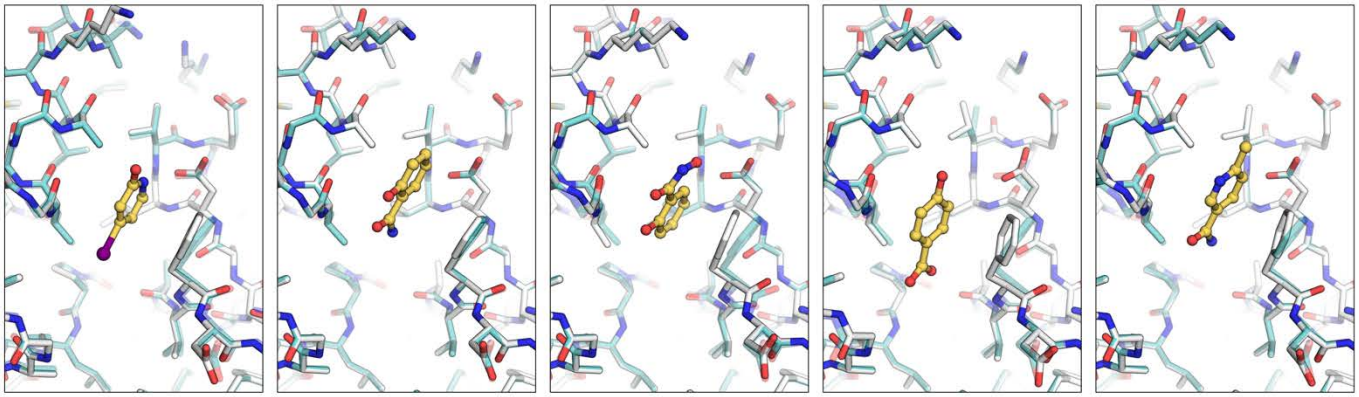


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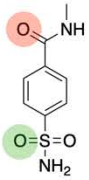


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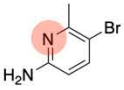
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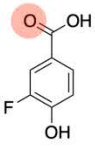
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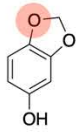
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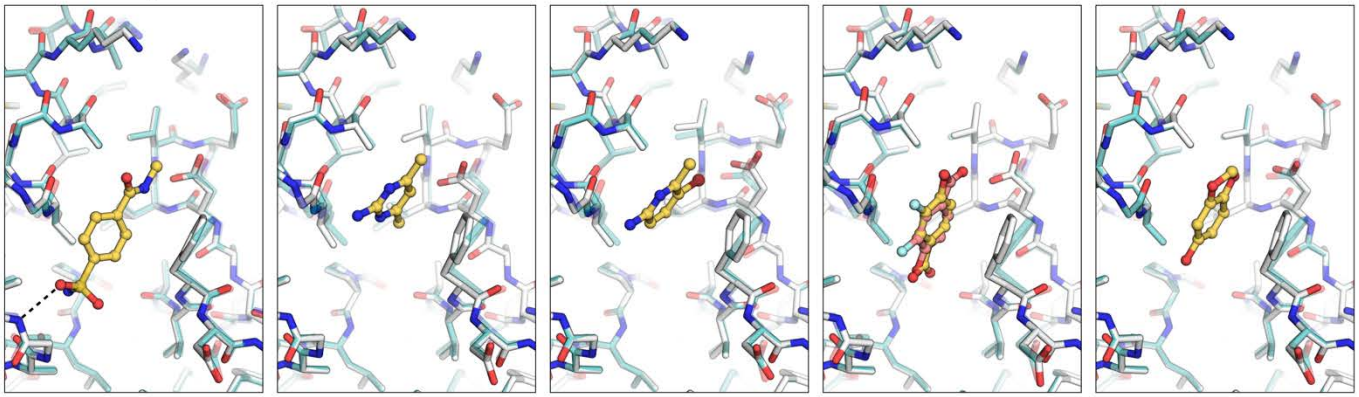


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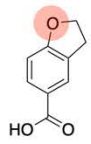


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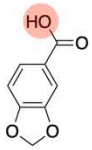




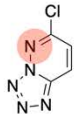
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UCSF-P0565_A | 5RV6
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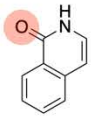
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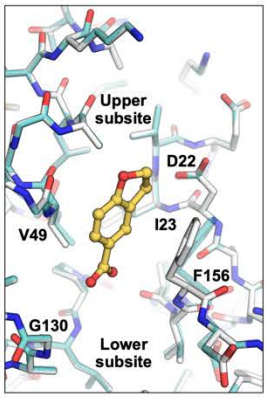


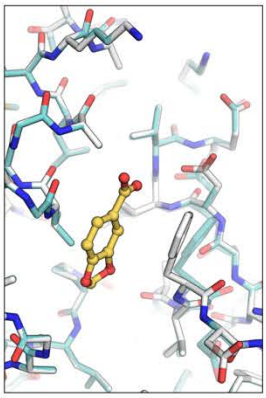
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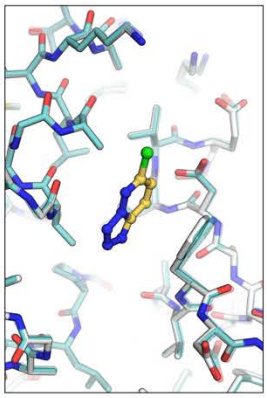


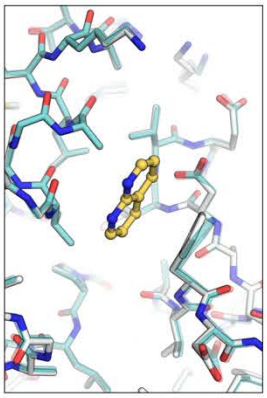
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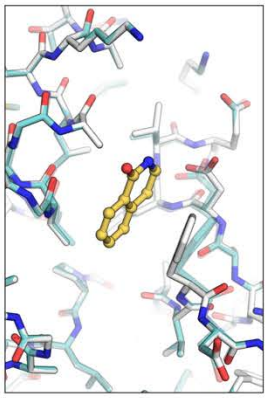




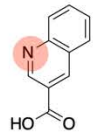




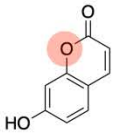




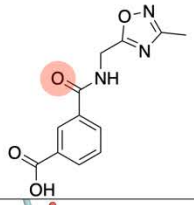
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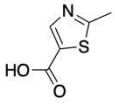
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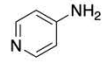
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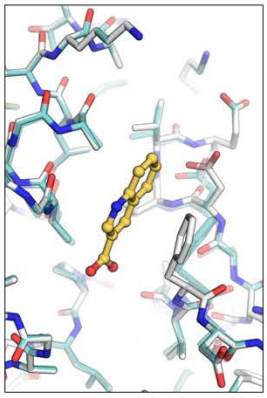


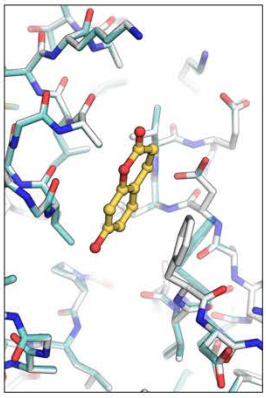
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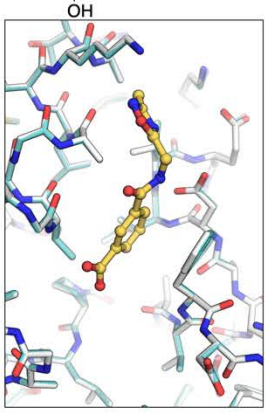


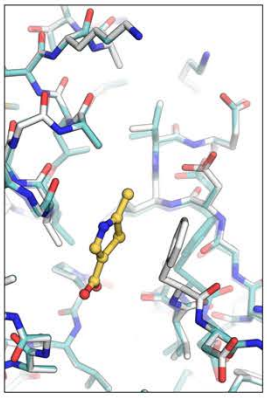
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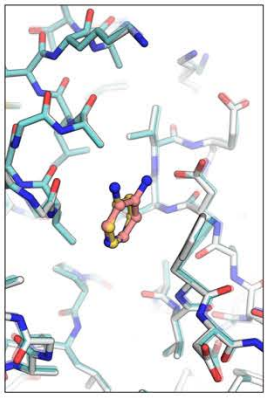




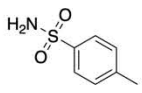




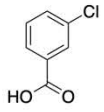




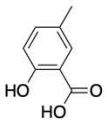
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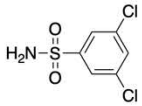
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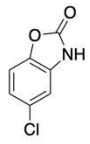
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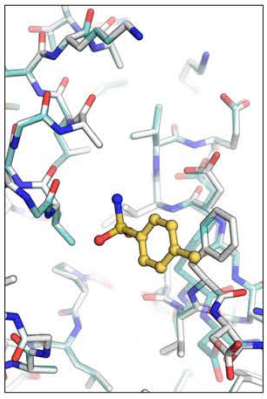


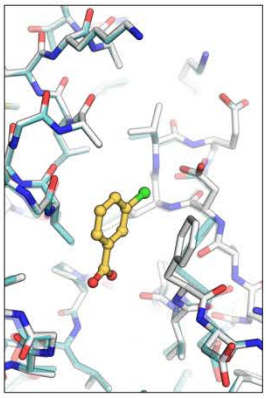
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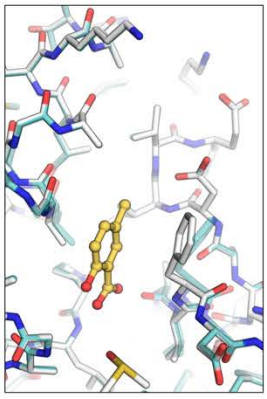


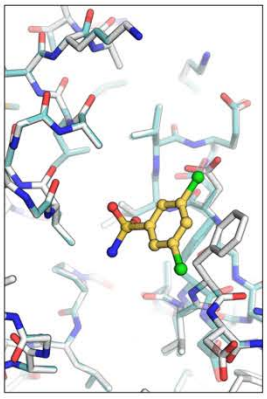
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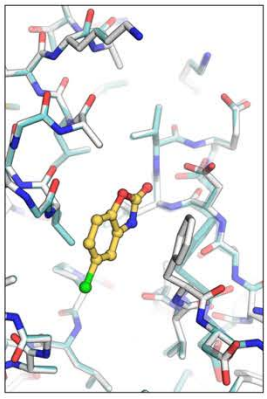




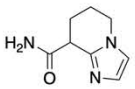




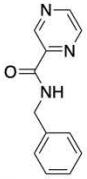




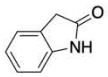
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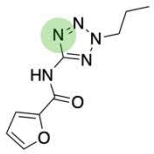
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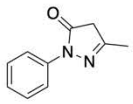
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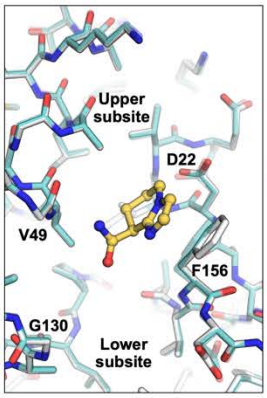


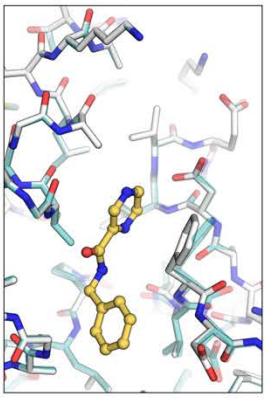
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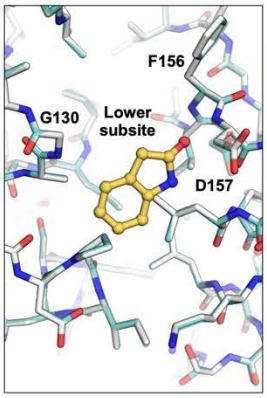


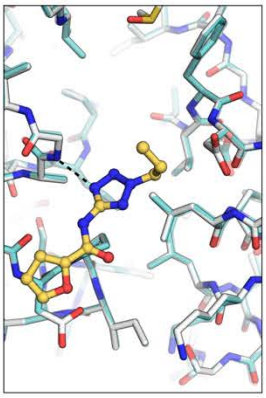
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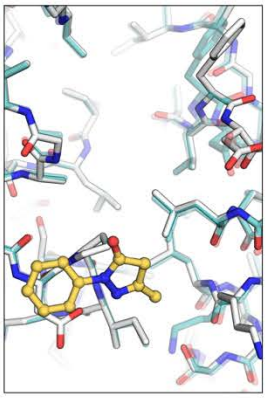




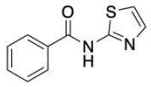




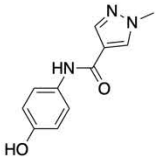




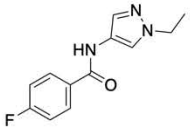
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ZINC39994



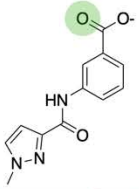
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Z1263529624



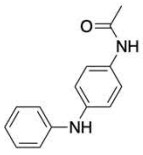
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Z373769142

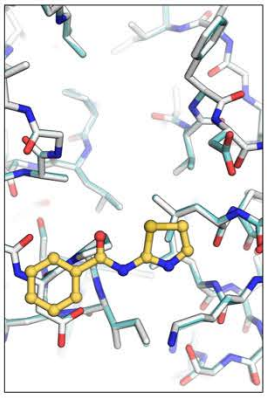


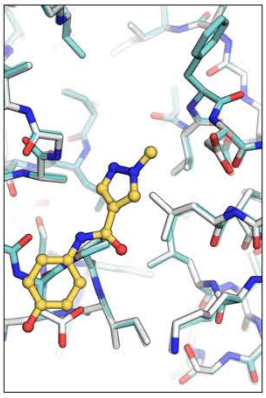
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Z2856434892

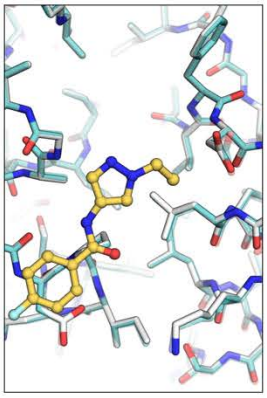


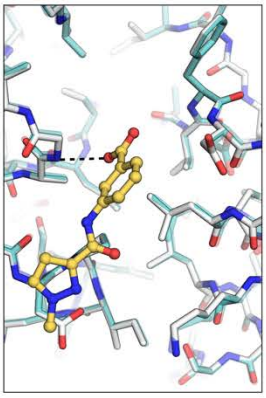
DLS-X0524_A | 5S35
Z68404778

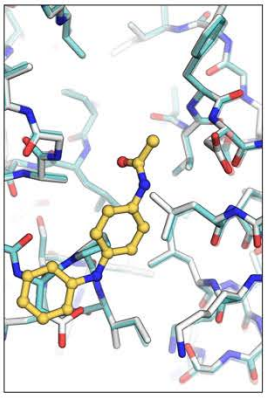




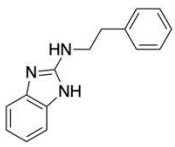


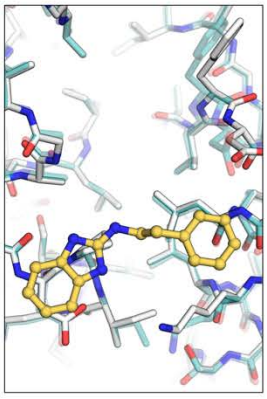






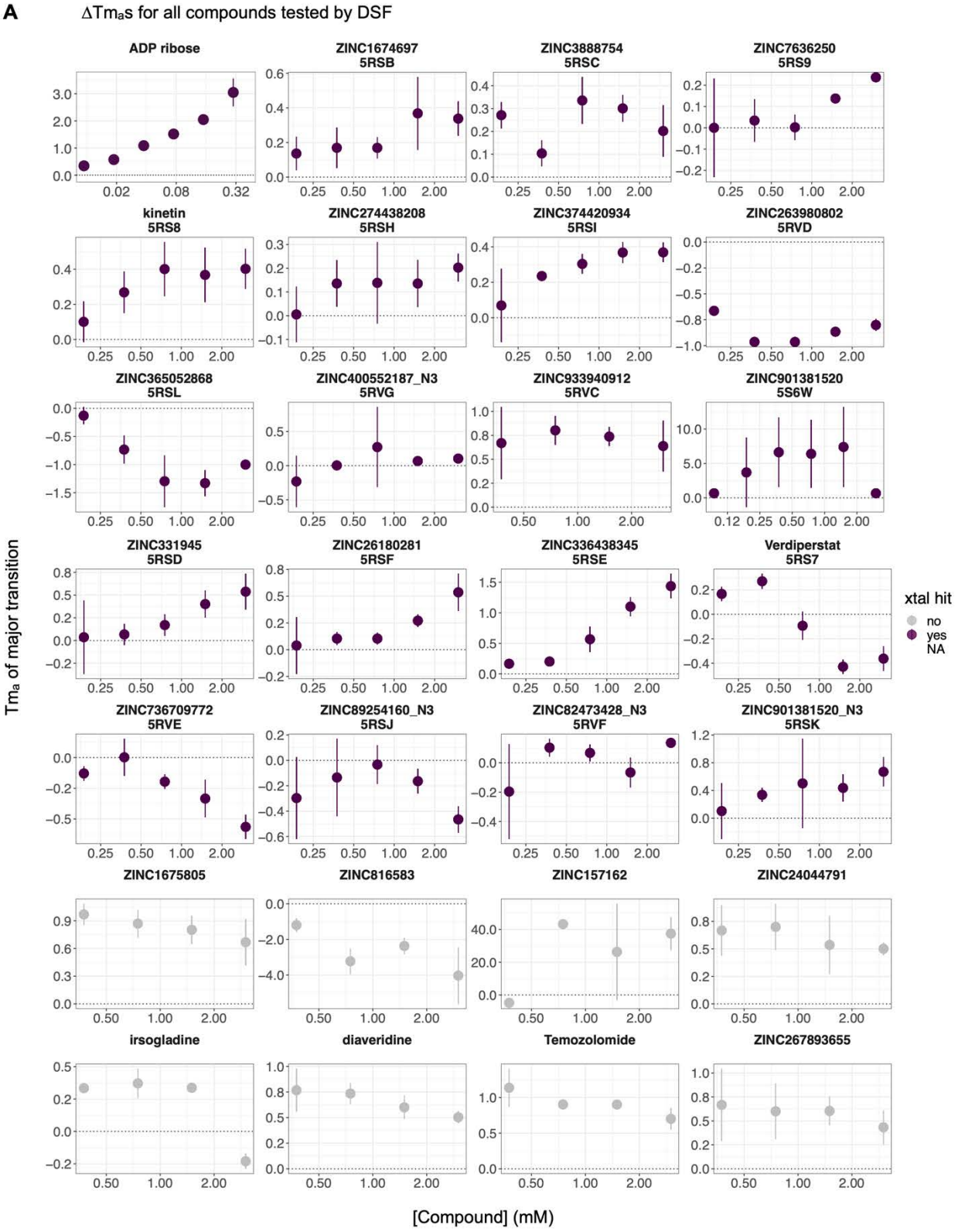
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Z57299529

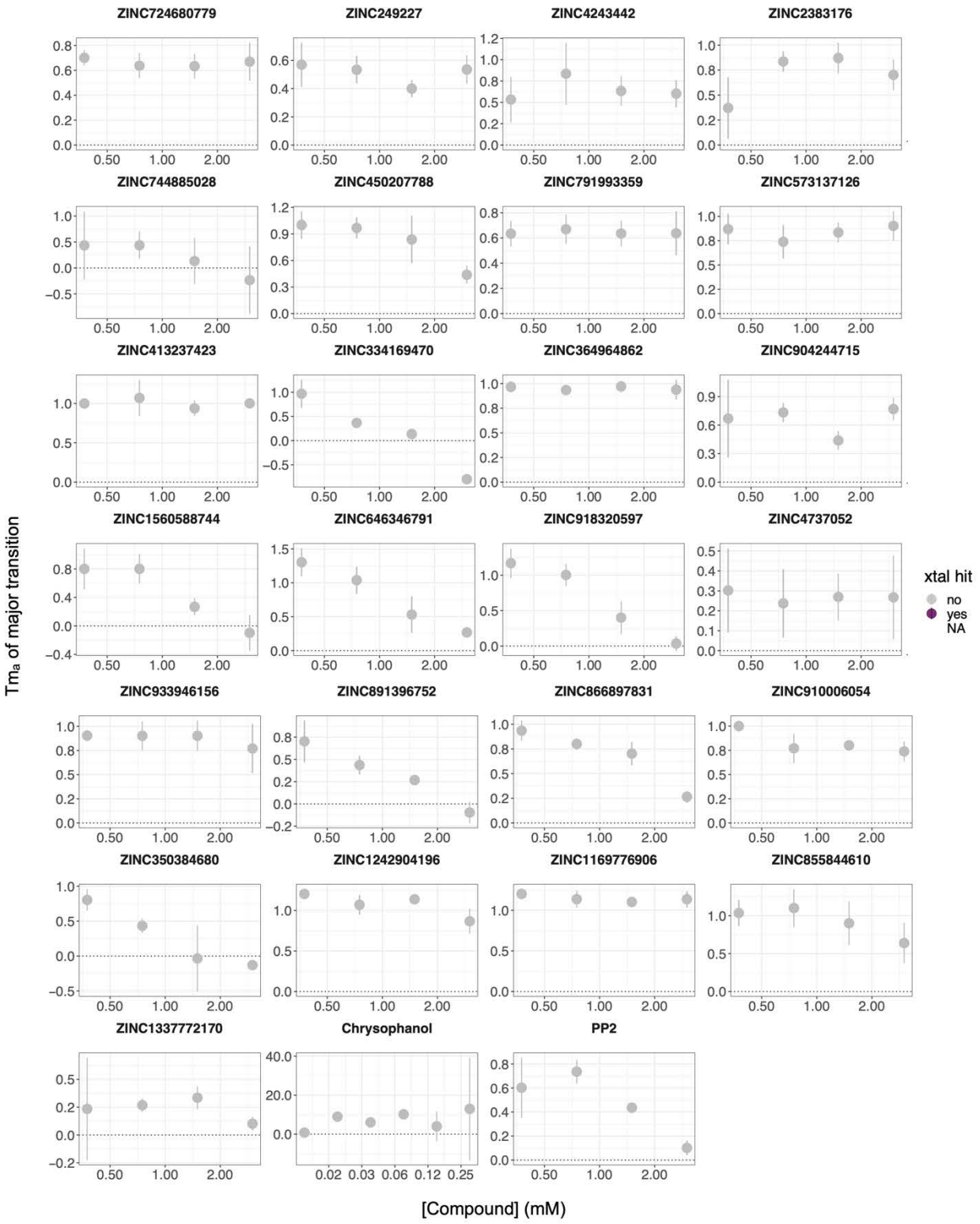




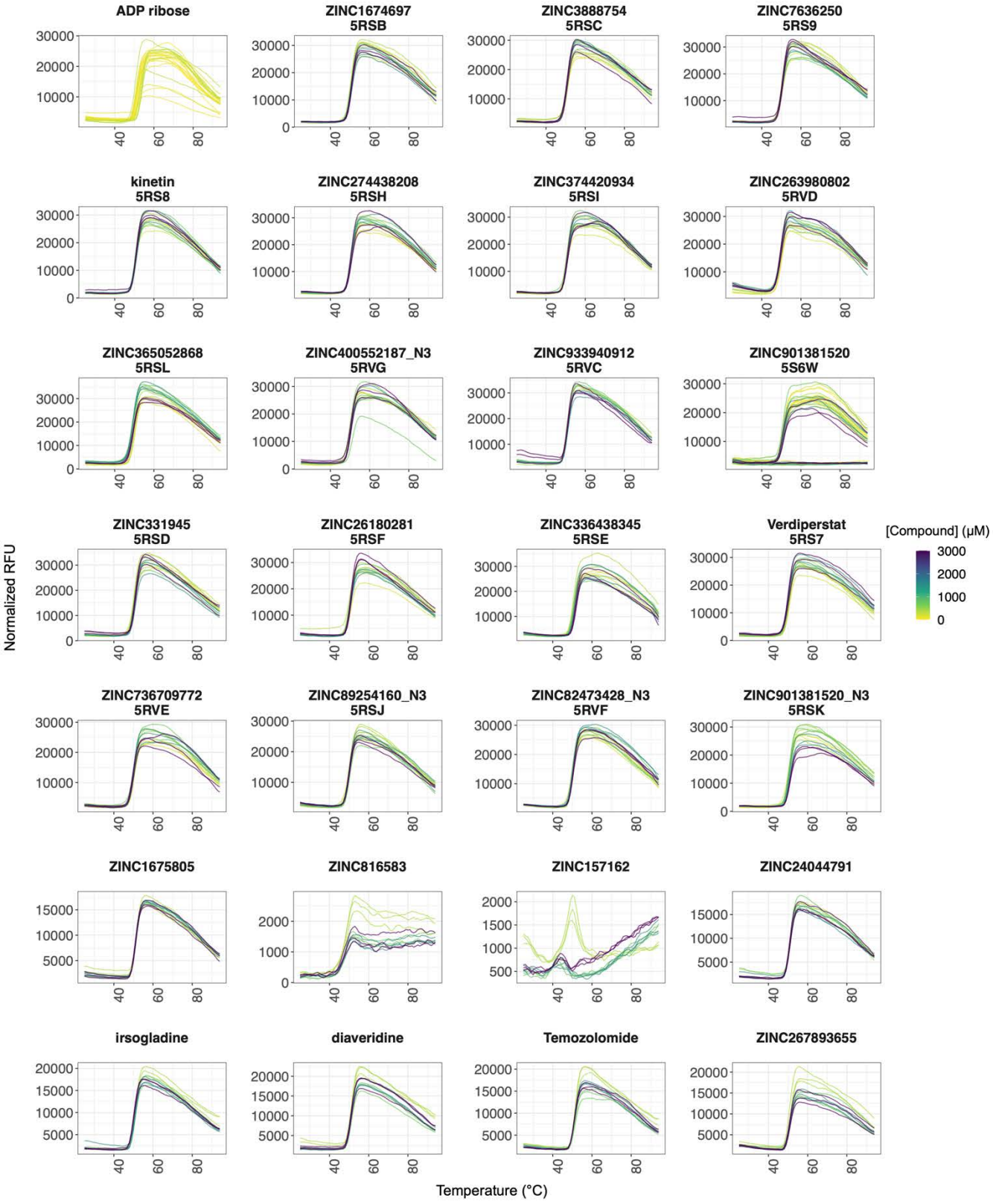
DSF results for all tested compounds

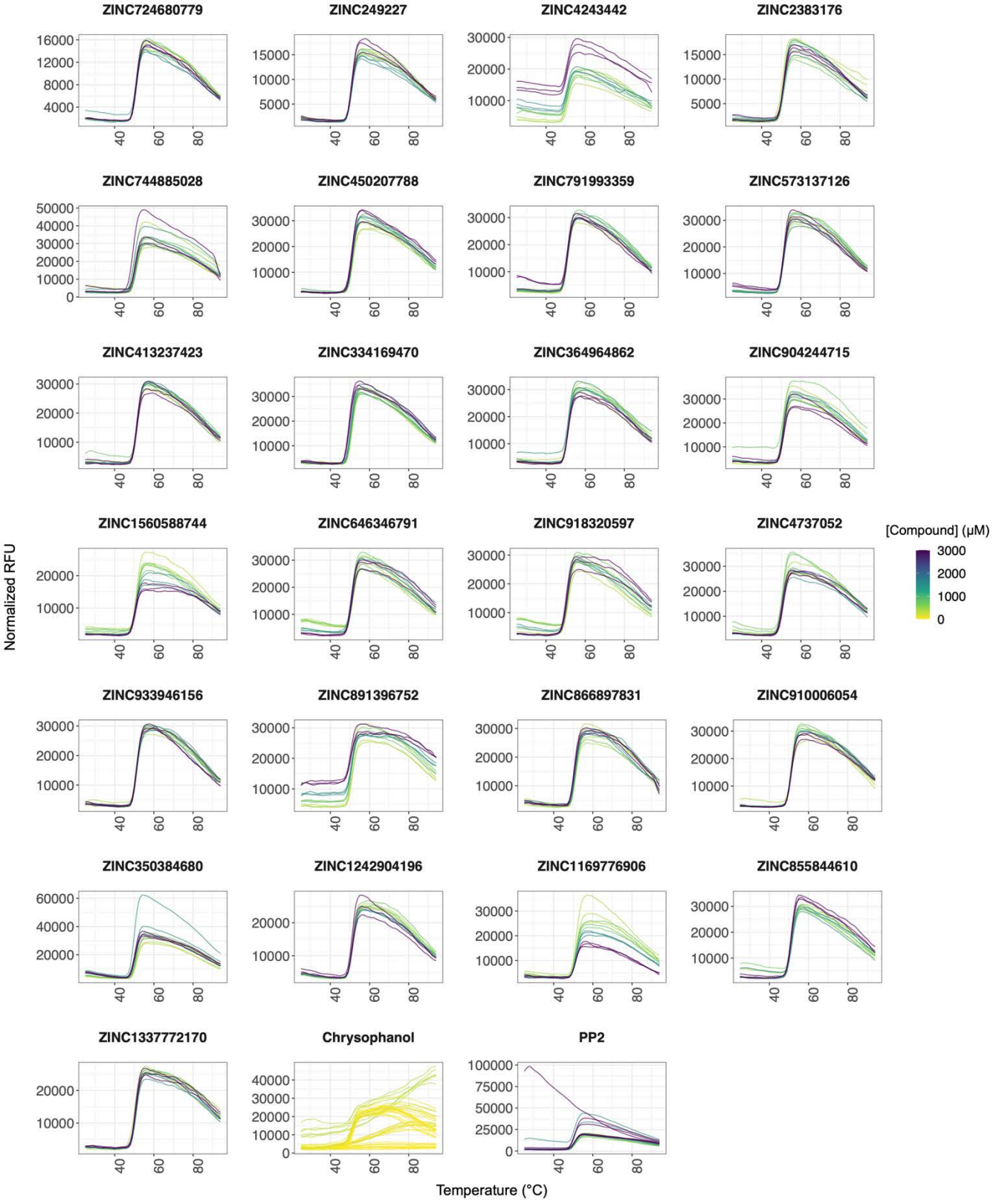
A) Changes in Tm_a observed in the presence of ADP-ribose (0-300 µM, positive control), as well as fragments (0-3 mM fragment) observed to bind by crystallography (purple), and those not observed to bind by crystallography (grey). Data points represent the mean ± SD for triplicate measurements at each compound concentration. **B)** Corresponding raw DSF data demonstrates the absence of confounding changes in curve shape for all compounds except ZINC157162.




ZINC724680779
ZINC249227
ZINC4243442
ZINC2383176
ZINC744885028
ZINC450207788
ZINC791993359
ZINC573137126
ZINC413237423
ZINC334169470
ZINC364964862
ZINC904244715
ZINC1560588744
ZINC646346791
ZINC918320597
ZINC4737052
ZINC933946156
ZINC891396752
ZINC866897831
ZINC910006054
ZINC350384680
ZINC1242904196
ZINC1169776906
ZINC855844610
ZINC1337772170
Chrysophanol
PP2
Tma of major transition
[Compound] (mM)
xtal hit
no
yes
NA

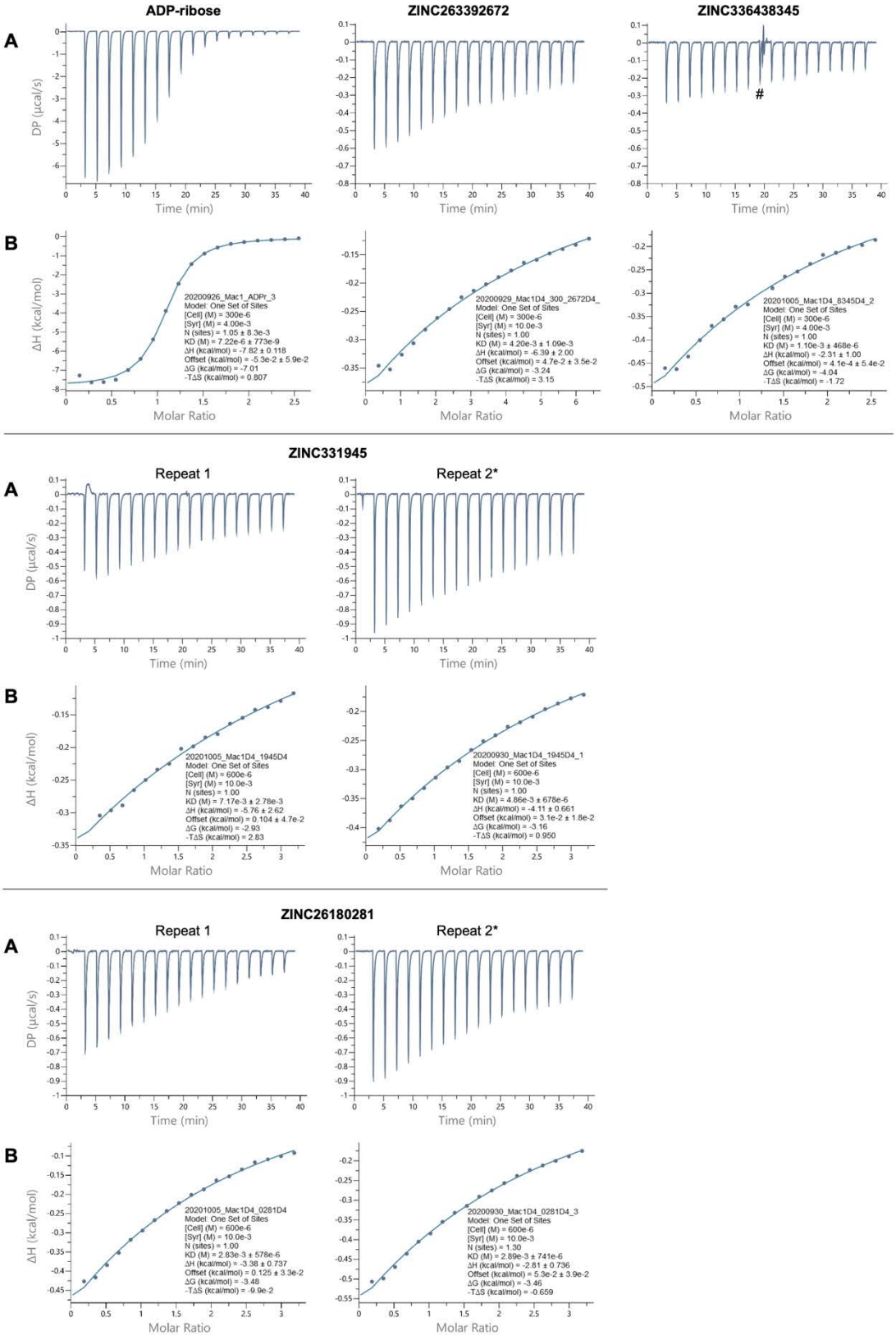
B Raw DSF data for all compounds tested by DSF

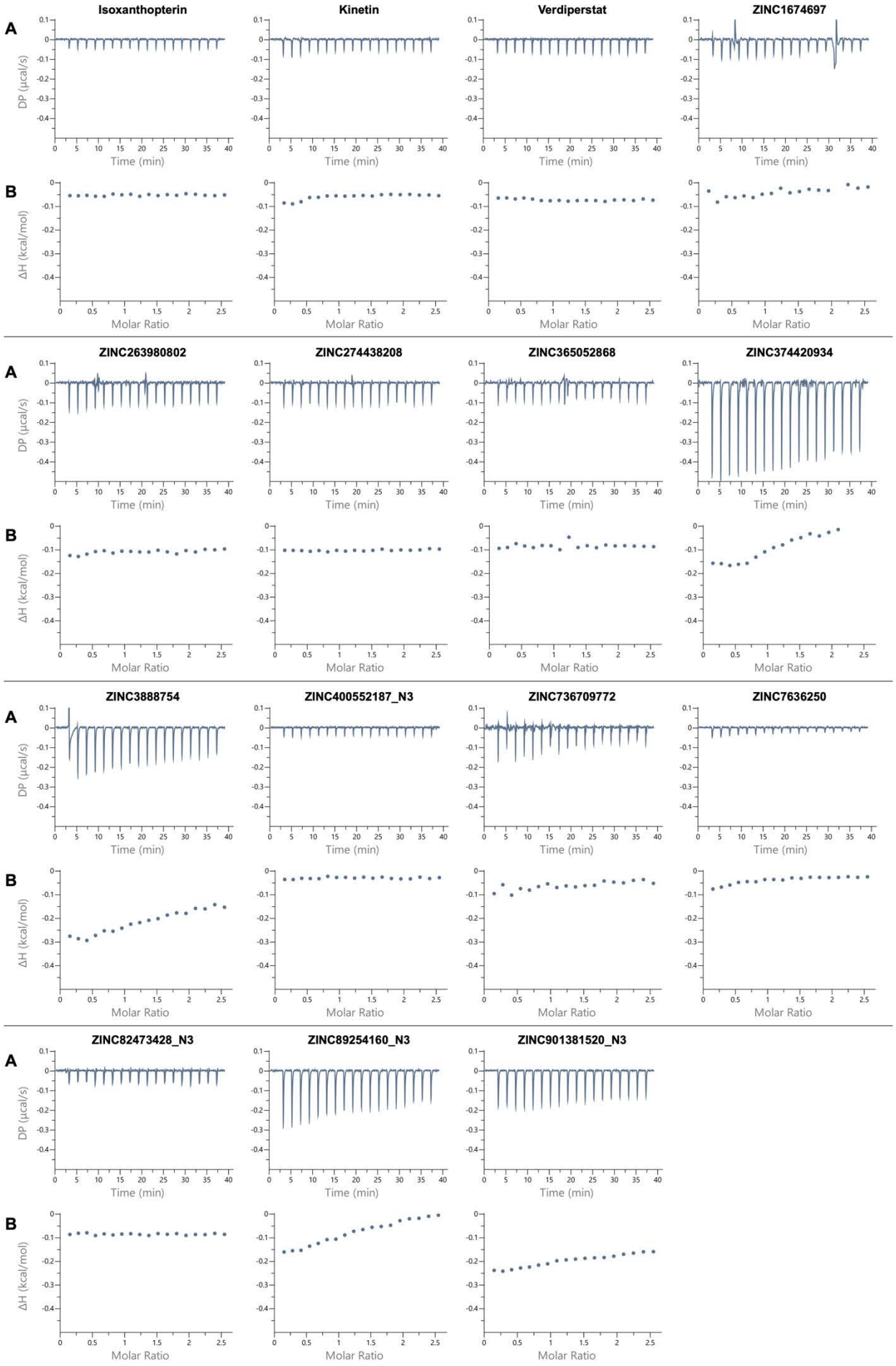



ZINC724680779
ZINC249227
ZINC4243442
ZINC2383176
ZINC744885028
ZINC450207788
ZINC791993359
ZINC573137126
ZINC413237423
ZINC334169470
ZINC364964862
ZINC904244715
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ZINC646346791
ZINC918320597
ZINC4737052
ZINC933946156
ZINC891396752
ZINC866897831
ZINC910006054
ZINC350384680
ZINC1242904196
ZINC1169776906
ZINC855844610
ZINC1337772170
Chrysophanol
PP2
Normalized RFU
Temperature (°C)
[Compound] (μM)
3000
2000
1000
0

ITC results for all tested compounds

A) Raw data for each fragment tested. ZINC26180281 and ZINC331945 were tested twice: the values reported in **Figure 9**E are from the second repeat (marked with *). One injection from the ZINC336438345 experiment was excluded from the analysis (marked with #). **B)** Binding isotherms for each fragment tested. Integrated heat peaks are represented as dots and the non-linear fit to a single-site binding model is represented as a line. The four fragments (and ADP-ribose) that could be fit using non-linear regression are shown first.

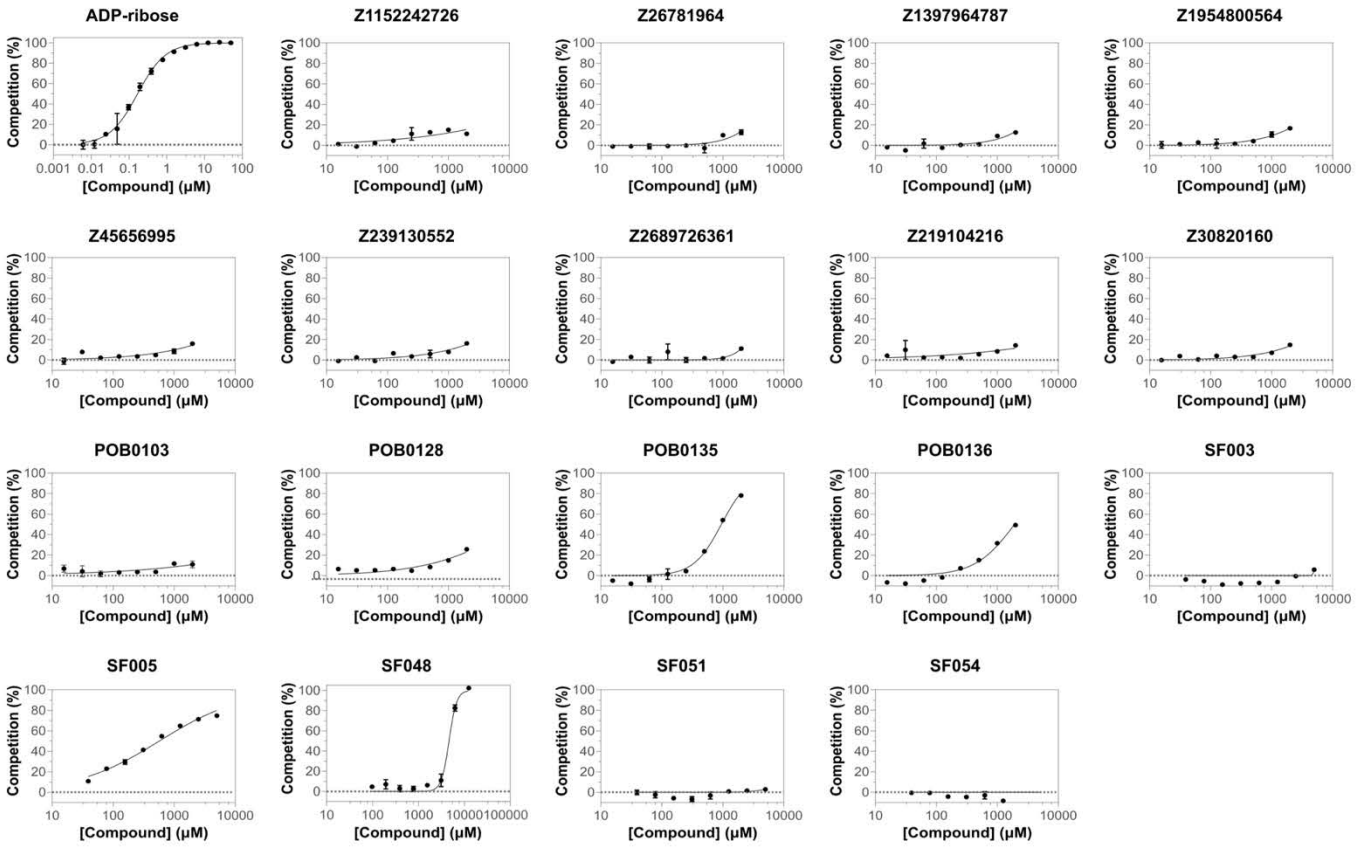



Isoxanthopterin
Kinetin
Verdiperstat
ZINC1674697
ZINC263980802
ZINC274438208
ZINC365052868
ZINC374420934
ZINC3888754
ZINC400552187_N3
ZINC736709772
ZINC7636250
ZINC82473428_N3
ZINC89254160_N3
ZINC901381520_N3
A
B
DP (µcal/s)
ΔH (kcal/mol)
Time (min)
Molar Ratio

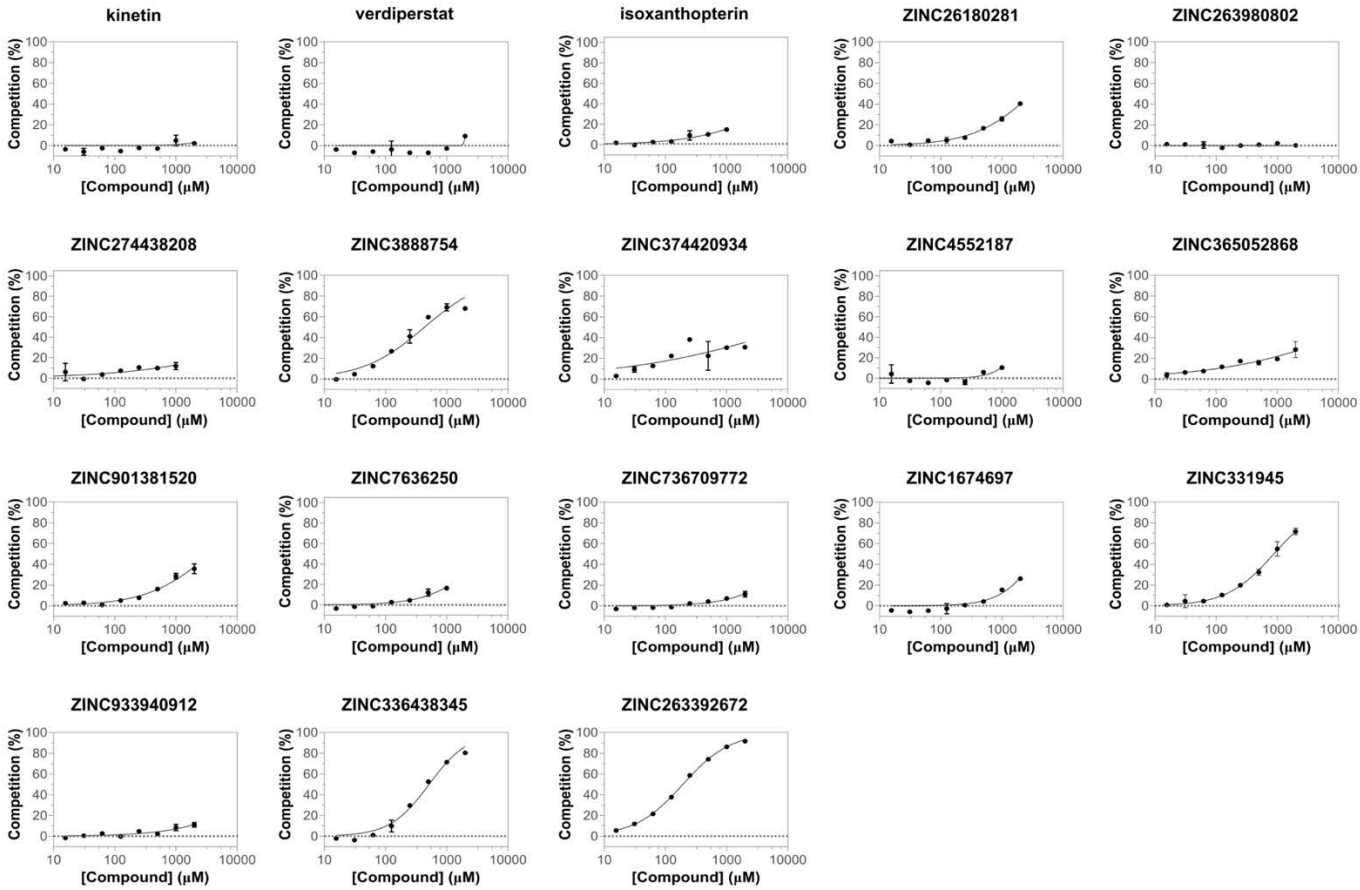
Macrodomain peptide displacement results for all tested compounds

A) Assay results obtained for selected crystallographic hits from the Oxford/XChem screen. **B)** Assay results obtained for docking hits that were crystallographically confirmed. **C)** Assay results obtained for docking hits that could not be crystallographically confirmed. ADP-ribose, used as a control compound in this assay, is shown in the upper left corner of **(A)**. All graphs show normalized compound competition (%) plotted against compound concentration with 1 mM or 2 mM top concentration. Non-linear fits of a sigmoidal dose-response equation (shown with a black line) were calculated for assay points (mean ± SEM, n = 2). Fits were constrained between 0 and 100% competition. IC_{50} results and competition (%) at the top concentration are summarized in **Data S1.**

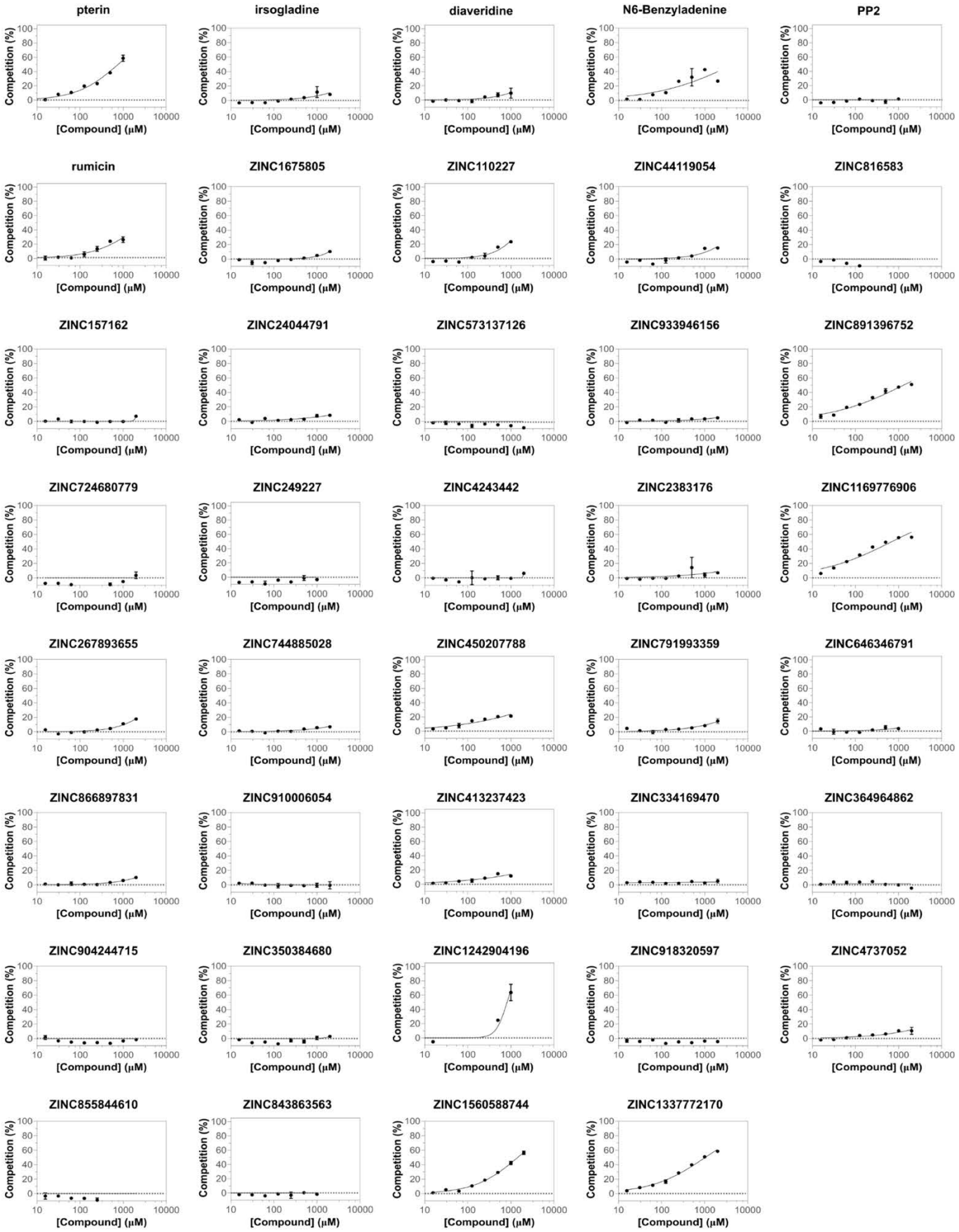
A



B


kinetin
verdiperstat
isoxanthopterin
ZINC26180281
ZINC263980802
ZINC274438208
ZINC3888754
ZINC374420934
ZINC4552187
ZINC365052868
ZINC901381520
ZINC7636250
ZINC736709772
ZINC1674697
ZINC331945
ZINC933940912
ZINC336438345
ZINC263392672
Competition (%)
[Compound] (μM)

c


pterin
irsogladine
diaveridine
N6-Benzyladenine
PP2
rumicin
ZINC1675805
ZINC110227
ZINC44119054
ZINC816583
ZINC157162
ZINC24044791
ZINC573137126
ZINC933946156
ZINC891396752
ZINC724680779
ZINC249227
ZINC4243442
ZINC2383176
ZINC1169776906
ZINC267893655
ZINC744885028
ZINC450207788
ZINC791993359
ZINC646346791
ZINC866897831
ZINC910006054
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ZINC364964862
ZINC904244715
ZINC350384680
ZINC1242904196
ZINC918320597
ZINC4737052
ZINC855844610
ZINC843863563
ZINC1560588744
ZINC1337772170
Competition (%)
[Compound] (μM)